

Asian Peptide Supplier Due-Diligence Dossiers

Combined Report

Contents

1. 06-M06.md
2. 07-M07.md
3. 08-M08.md
4. 09-M09.md
5. 10-M10.md

06-M06.md

M06 — Shilpa Pharma Lifesciences / Shilpa Medicare (India)

****Decision date:**** 20 August 2026

****Purpose:**** Preliminary desk prequalification for regulated therapeutic-peptide supply to a South African business

****Preliminary desk tier:**** ****B** — credible, with material gaps requiring a formal RFI and site-specific audit evidence

1. Executive conclusion

Shilpa presents ****credible regulated-market evidence for chemically synthesised peptide active pharmaceutical ingredients (APIs/drug substances)****. The strongest public evidence is: (i) the audited parent filing identifies Shilpa Pharma Lifesciences Limited (“SPLL”) as the wholly owned API subsidiary operating two Raichur API sites; (ii) parent stock-exchange disclosures report EDQM Certificates of Suitability (“CEPs”) for the synthetic peptide APIs ****desmopressin**** and ****octreotide****, both made by solid-phase synthesis; and (iii) FDA inspected both Raichur API units in March 2025, with Unit I ultimately reported as ****VAI**** and Unit II reported by the company as having no Form 483 observations.[1][4][5][6] Unit II also has company-filed evidence of PMDA and ANVISA API inspections.[7][8]

This is not yet enough for Tier A. Public sources do not unambiguously tie the peptide trains, each peptide CEP/DMF, preparative purification equipment, lyophilizers, warehouses, and release laboratories to ****one named API unit and building****. The marketed annual peptide drug-substance capacity of ****40 kg**** and cartridge/device fill-finish capacity of ****20 million units**** originates from a Shilpa press release relayed by trade media, without a named site, line, equipment list, inspection scope, or regulator certificate.[16] Public evidence also does not establish a South African peptide-product registration, a SAHPRA-recognised peptide API source, or a South African customer. The parent’s Jadcherla finished-dose site has serious historical FDA findings and remained listed in the current FDA Import Alert 66-40 source retrieved for this review; that history is ****site- and operation-specific to finished pharmaceuticals****, not proof of a peptide API defect, but it is material to any proposal involving group fill-finish, sterile vials, packaging, or shared quality governance.[10][11]

Accordingly, SPLL is ****suitable to advance to an RFI for peptide API/drug substance****, particularly desmopressin or octreotide, but sterile drug product, cartridge/device, and integrated drug-substance-to-drug-product proposals should remain conditional on documentary closure of the Jadcherla FDA history and identification/audit of the actual peptide and fill-finish lines. No finding in this dossier should be read as saying the company is “fully compliant,” “SAHPRA-approved,” or “safe.”

2. Legal entity and corporate control

The relevant contracting/manufacturing entity for API work is ****Shilpa Pharma Lifesciences Limited****, an Indian public limited company reported under CIN ****U24100KA2020PLC134081**** in public corporate-directory records. The listed parent is ****Shilpa Medicare Limited****, CIN ****L85110KA1987PLC008739****, headquartered at Shilpa House, #12-6-214/A1, Hyderabad Road, Raichur 584135, Karnataka.[1][2][3]

The audited FY2025–26 consolidated financial statements list SPLL as an Indian subsidiary held ****100.00%**** by Shilpa Medicare Limited. The report says the API business is operated through SPLL and encompasses oncology and non-oncology APIs, intermediates, peptides, polymers, payloads, linkers and CDMO services.[1] This legal distinction matters: peptide API quality agreements, change notification, pharmacopoeial commitments, CEP/DMF access letters and batch release should name ****SPLL and the exact registered manufacturing site****, while any

separate finished-dose or analytical contract may name Shilpa Medicare Limited or another group company.

3. Relevant sites and operational boundaries

| Legal entity / site | Publicly evidenced role | Therapeutic-peptide relevance and limitations |

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|---|

| ****SPLL Unit I****, Plot No. 4A (the annual report lists a wider DTA plot range), Deosugur Industrial Area, Deosugur/Shaktinagar 584170, Raichur District, Karnataka | API manufacture. FDA inspection 3–7 March 2025; company reported EIR and ****VAI**** classification. The group location page identifies it as Raichur API Unit I.[1][3][6] | Potential peptide API site, but public sources reviewed do not assign a specific peptide, CEP, SPPS suite, preparative HPLC or lyophilizer to Unit I. VAI warrants the underlying Form 483, response, EIR and CAPA-effectiveness package in RFI. |
| ****SPLL Unit II / 100% EOU and R&D Centre****, Plot Nos. 33, 33A and 40–47, Raichur Industrial Growth Centre, Wadloor Road, Chicksugur 584134, Karnataka | API manufacture and R&D. Company reported FDA inspection 10–14 March 2025 with no Form 483 observations; PMDA inspection 29 August–1 September 2023 without critical/major observations; ANVISA inspection 15–19 April 2024 followed by a GMP certificate.[3][7][8] | The current Shilpa Pharma site presents this address as its manufacturing site and describes dedicated peptide production/purification/lyophilisation infrastructure.[12] This makes Unit II the more likely peptide locus, but that inference is ****not accepted as proof****. RFI must confirm each peptide and CEP/DMF site linkage. |

| ****Shilpa Medicare Limited Unit IV, Jadcherla****, Plot S-20 to S-26, Pharmaceutical Formulations SEZ, TSIC Green Industrial Park, Polepally/Pollepally Village, Jadcherla Mandal, Mahbubnagar 509301, Telangana | Sterile injections and non-sterile tablets/capsules; parent report states dry-powder and liquid injectable/lyophilisation capability, approximately 3 million liquid vials and 2 million lyophilised vials.[1][9][10] | ****Finished dose/fill-finish only; not evidenced as peptide API manufacture.**** AGES issued EU GMP after a January 2024 inspection according to the company.[9] FDA issued a 2020 warning letter for serious finished-pharmaceutical CGMP violations and the site was placed on Import Alert 66-40.[10][11] Do not include this site in a peptide supply chain without current FDA status, line-specific aseptic qualification and regulator scope. |

| ****Shilpa Medicare Unit VII****, Plot 79, Road 15, Survey 125, IDA Mallapur/Nacharam, Hyderabad 500076, Telangana | Analytical services and bioanalytical laboratory according to the location page.[3] | Could be a subcontracted or group release/stability laboratory. Public sources do not show which peptide methods or batches are tested there. All testing flows, sample custody, raw-data systems, data-review responsibility and quality-agreement controls must be declared. |

| ****Shilpa Medicare Unit VI, Dabaspur/Bengaluru****, Plot 29/A5, Phase IV, Averahalli/Sompura Industrial Area, Bengaluru Rural 562111, Karnataka | Oral dissolving films and transdermal patches in the annual report.[1] | Not an injectable-peptide or peptide-API site on available evidence. It should not be represented as peptide fill-finish or device assembly without new documentary evidence. |

| ****Shilpa Biologicals / Dharwad group facility**** | Biologics drug substance, drug product, prefilled syringes and packaging are described in the annual report.[1] | Separate modality and legal entity. Public materials do not prove that chemically synthesised peptide drug substance is filled here. Exclude from therapeutic-peptide qualification unless a proposed product specifically uses this site and the site's authorisation covers it. |

4. Capability classification

4.1 Peptide API / drug substance — credible but partly marketing-derived

Two product-level regulatory milestones are verified through dated parent-company stock-exchange disclosures:

* ****Desmopressin:**** SPLL announced an EDQM CEP on 3 October 2024 and explicitly described

desmopressin as a synthetic peptide made through **solid-phase synthesis**. The current product page gives CEP **2024-028 Rev 00**, but the regulator database record and exact ma

nufacturing unit were not independently extracted in this review.[4][13]

* **Octreotide**: SPLL announced an EDQM CEP on 18 November 2024 and likewise described solid-phase synthesis. The current product page gives CEP **2024-102 Rev 00**, again without an independently captured EDQM database record tying it to a unit.[5][13]

* **Liraglutide**: the current company product page states US DMF **043506**. A DMF number is evidence of a regulatory submission mechanism, not approval of a finished product, proof that the DMF is active/adequate, or permission to market.[13]

* **Semaglutide**: the company page says “process validated.” This is a supplier assertion; no public CEP, approved-product linkage, FDA DMF status or regulator inspection scope specific to semaglutide was verified.[13]

* **Tirzepatide, dulaglutide, sermorelin, teduglutide and tesamorelin**: the company itself labels these as laboratory development/validation in progress. They are not treated as qualified commercial therapeutic supply.[13]

Shilpa markets SPPS (Fmoc and Boc), solution/liquid-phase synthesis, hybrid SPPS–LPPS, fragment condensation, native chemical ligation and recombinant/fermentation routes. It also markets lipidation, PEGylation, cyclisation, disulfide folding, unnatural amino-acid incorporation and other modifications.[14] The legacy site says it has multiple SPPS reactors, a microwave synthesiser, preparative HPLC, lyophilisation equipment and peptide precipitation know-how, and that GMP desmopressin can be supplied from mg to kg scale.[15] The newer corporate capability page describes dedicated R&D/production blocks, preparative HPLC/column purification, lyophilizers, cold rooms and medium-to-commercial supply, but does not publish reactor resin loading, vessel volume, preparative-column dimensions, lyophilizer shelf area or routine validated batch size.[12]

The trade press relays Shilpa's claim of **40 kg/year peptide drug-substance capacity**. [16] Because this figure lacks a named unit, qualified equipment train, utilisation basis and independent audit evidence, it should be treated as **marketing capacity**, not confirmed available or validated capacity.

4.2 Sterile finished dose and fill-finish — separate and not yet peptide-qualified

The audited report evidences Jadcherla capability for oncology tablets, capsules and injectables, including dry powder and liquid lyophilisation, at approximately 3 million liquid vials and 2 million lyophilised vials.[1] It does not identify an approved peptide presentation produced there. The 2025 hybrid-CDMO announcement, relayed by trade media, claims cartridge and device fill-finish capacity of **20 million units/year**, but provides no site, dosage-form authorisation, line description, isolator/RABS status, device constituent, regulator inspection or approved peptide label.[16]

Therefore:

* **Vial filling and finished-dose lyophilisation** are evidenced as group capability at Jadcherla, but peptide-specific GMP readiness is unverified.

* **Cartridge filling, prefilled syringe filling, pen/autoinjector assembly or combination-product manufacture** are not accepted for therapeutic qualification from the public evidence reviewed.

* **Packaging/device services** are a marketing claim. No public evidence reviewed identifies device suppliers, assembly location, ISO 13485 scope, design-responsibility allocation, human-factors capability, container-closure validation or South African device establishment status.

4.3 Research-only and non-GMP services — excluded

The legacy site explicitly offers **non-GMP/GLP-grade** peptide and analytical services from mg-to-kg laboratory scale, stating that non-GMP and GMP facilities reside “at the same place.”[17]

Shilpa also markets cosmetic and speciality peptides, and the current catalogue includes products such as BPC-157, Selank and TB-500 alongside therapeutic APIs.[13] These catalogue/research/cosmetic offerings must be segregated from a therapeutic

supply qualification. No non-GMP material, catalogue certificate, research-only sample or grey-market product should enter development, validation, clinical or commercial therapeutic supply without a formally controlled GMP technology-transfer and comparability pathway.

5. Regulatory evidence and market mapping

| Market / authority | Evidence | Qualification interpretation |

|---|---|---|

| ****European Pharmacopoeia / Europe**** | Parent disclosures report EDQM CEPs for desmopressin and octreotide; AGES inspection and EU GMP certificate were reported for Jadcherla in 2024.[4][5][9] | Strongest peptide-specific evidence is the two API CEPs. A CEP supports use of the substance monograph and covered manufacturing route; it is not a marketing authorisation for a peptide medicinal product. Jadcherla's EU GMP evidence is finished-dose/site-specific and does not prove peptide fill-finish. |

| ****United States**** | FDA inspected API Unit I and Unit II in March 2025. Unit I was reported VAI with EIR; Unit II was reported zero Form 483 observations. Company states liraglutide DMF 043506. FDA's 2020 warning letter and current Import Alert 66-40 concern Jadcherla Unit IV finished pharmaceuticals.[6][10][11][13] | Supports RFI-stage confidence in API regulatory exposure, but Unit I CAPA and DMF status must be reviewed. Do not extrapolate Unit II's inspection outcome to Unit I or Jadcherla. No US-approved peptide product linked to SPLL was verified. |

| ****Japan**** | Company stock-exchange filing reports PMDA inspection of Unit II's oncology/non-oncology API facility in Aug–Sep 2023, concluding without critical/major observations.[7] | Credible site-level API exposure, but public filing does not say peptides were in scope or identify a Japanese approved peptide product. |

| ****Brazil**** | Company stock-exchange filing reports ANVISA inspection of Unit II in April 2024 and issuance of a GMP certificate for the API facility.[8] | Credible API-site evidence; peptide scope and current certificate should be obtained. No Brazilian peptide approval linked to SPLL was verified. |

| ****Australia, Korea, Mexico, Canada and WHO**** | Corporate pages/annual report claim TGA, KFDA/MFDS, COFEPRIS, Health Canada and WHO-GMP accreditations across group facilities.[1][2][14] | Site, product, activity and validity are insufficiently resolved for peptide qualification. Treat as unverified until certificates/database extracts are supplied. |

| ****South Africa**** | The FY2025–26 annual report prints "SHAPRA" among Jadcherla accreditations and the marketing site lists South Africa among markets served.[1][12] | This is not accepted as evidence of a peptide product registration, SAHPRA GMP status, a registered API source, or supply history. No SAHPRA database record or South African product label naming SPLL/Shilpa as peptide manufacturer was verified. ****South African therapeutic market service is not publicly disclosed on reliable product-level evidence.**** |

The audited report gives group-level FDF filings and revenues in the US, Europe, Canada, India and rest-of-world, while the API narrative refers to customers in the US, Europe and Japan.[1] Those statements establish general market reach, not peptide customers or peptide product approvals. No customer is inferred in this dossier.

6. Quality control, analytical capability and peptide impurity control

Shilpa's current marketing materials describe the following:

* HPLC, UPLC, GC and LC–MS for assay, related substances, residual solvents and impurity profiling; XRD, DSC and spectroscopy for characterisation; microbiology labs; and ICH stability chambers.[12]

* LC–MS/GC–MS identification of unknown impurities; nitrosamine, non-carryover, genotoxic and elemental impurity testing; qualification of impurity/reference/working standards; pharmacopoeial

testing; HPLC/GC method development and validation; cleaning methods; forced degradation; and method transfer.[18]

* Stability storage at **–20°C, 2–8°C, 25°C/60% RH, 30°

C/65% RH and 40°C/75% RH**.[18]

* Peptide-process development addressing deletion, truncation, insertion sequences, D-isomers, aggregation, counter-ion exchange and TFA removal; sequence confirmation by RP-HPLC, LC–MS and peptide mapping; lyophilisation-cycle development; QbD/DoE; and annual product reviews.[14]

These are useful RFI leads, not independently verified execution records. For desmopressin, the legacy page states a GMP batch was released to USP, EP and JP specifications.[15] Public sources do not provide representative certificates of analysis, chromatograms, mass spectra, method-validation summaries, impurity fate/purge studies, residual-TFA limits, peptide content/water correction, amino-acid analysis, optical purity methods, endotoxin/bioburden limits, elemental/nitrosamine risk assessments or stability trends.

OOS/OOT, CAPA and data integrity

The most direct public quality-system evidence is adverse and belongs to **Shilpa Medicare Unit IV, Jadcherla**, not SPLL's API sites. FDA's 2020 warning letter found inadequate OOS and complaint investigations, unsupported attribution of OOS results to analytical error, repeated column reconditioning/retesting, failure to investigate an unknown HPLC peak adequately, weak CAPA detail/effectiveness checks, release after assay OOS retesting, a vial-seal complaint linked to sealing-machine defects, and a late Field Alert Report.[10] FDA explicitly said the response was inadequate in the discussed areas. This is material to group governance and any use of Jadcherla, but it must not be falsely described as a peptide API observation or automatically transferred to SPLL.

No public, peptide-site-specific evidence was found for:

- * electronic raw-data architecture, audit-trail review, user-access administration or ALCOA+ controls;
- * OOT procedure and statistical trending;
- * peptide-specific laboratory-investigation procedure and retest/resample rules;
- * CAPA ageing and effectiveness metrics;
- * deviation recurrence, right-first-time rate, invalidated assay rate or batch rejection history;
- * change control across CEP/DMF commitments;
- * current quality agreements governing Unit VII or third-party laboratories.

These are mandatory RFI and audit topics. Unit I's 2025 VAI classification adds urgency to obtaining the Form 483, response, EIR, CAPA and evidence of effectiveness.[6]

7. Adverse regulatory history

FDA Warning Letter 320-21-01 / MARCS-CMS 607877 (9 October 2020): FDA inspected Unit IV, Jadcherla in February 2020 and described significant finished-pharmaceutical CGMP violations. The site manufactured sterile injectables and solid dosage products. Findings included inadequate OOS and complaint investigations/CAPA, an inadequately investigated chromatographic impurity/unknown peak, release after unsupported retesting, vial sealing/closure concerns and failure to submit a timely Field Alert Report.[10]

FDA Import Alert 66-40: the FDA current page retrieved for this decision cycle was published 17 August 2026 and search results continued to identify Shilpa Medicare Limited at the Jadcherla address, originally dated 16 February 2021, under detention without physical examination for drug-GMP concerns.[11] Historical company/media reporting says azacitidine injection, cyclophosphamide capsules and erlotinib tablets were excluded; exclusions must not be assumed to cover any peptide. Before relying on Jadcherla, obtain the live red-list extract, any

close-out/removal letter, latest inspection classification, all 2025 inspection records and product-level admissibility advice. The FDA warning-letter page reviewed did not display a close-out letter.

****API sites:**** no FDA warning letter or import alert specific to SPLC Unit I or Unit II was found in the sources reviewed. Unit I's March 2025 inspection was VAI, whereas Unit II was reported to have no Form 483 observations.[6] The distinction is essential: favorable Unit II evidence cannot erase Unit I's VAI or Unit IV'

s enforcement history.

No peptide-specific recalls, WHO notices of concern, EDQM suspension, TGA recall, Health Canada recall, PMDA administrative action, NMPA action or MFDS action were verified in this desk review. Absence from public search results is not proof that no event exists.

8. Supply-chain and concentration risks

8.1 Raw materials and solvents

SPPS and hybrid peptide production inherently depend on qualified protected amino acids, coupling reagents, resins/linkers, cleavage reagents and large solvent volumes. Shilpa markets resin/linker and Fmoc/Boc strategy optimisation but does not identify suppliers, countries of origin, dual-source status, lead times, specifications, change-notification controls or inventories.[14] The audited parent report discloses foreign-currency expenditure for imported raw materials and, at group level, identifies dependence on globally sourced raw materials amid geopolitical and logistical challenges as a threat.[1]

RFI must obtain approved-vendor lists and country-of-origin mapping for at least protected/non-natural amino acids, peptide resin, fatty-acid sidechains/linkers for GLP-1 products, coupling reagents, TFA or alternative cleavage reagents, acetonitrile, DMF/NMP and other process solvents, counter-ions, primary standards and chromatographic media. Particular attention is required for single-source unnatural amino acids, long-lead resins, high-purity acetonitrile and regulated residual-solvent recovery/disposal.

8.2 Site and equipment concentration

Public sources suggest peptide development and GMP operations are in Raichur and may be co-located at Unit II, but do not show a qualified alternative train or second approved peptide site.[3][12][17] A CEP/DMF tied to one unit, one preparative column or one lyophilizer can create single-site and single-equipment exposure. Obtain business-continuity plans, spare-column strategy, lyophilizer redundancy, utility/water redundancy, disaster recovery, maximum tolerable downtime and alternate-release-laboratory plans.

8.3 Purification and lyophilisation bottlenecks

Preparative HPLC resin/media availability, solvent supply/recovery, fraction pooling analytics and lyophilizer capacity can constrain output independently of reactor capacity. The marketed 40 kg/year headline does not establish purified, released, saleable annual output.[16] Capacity should be evaluated using sequence-specific yield, cycle time, chromatography loading, number of purification passes, lyophilizer occupancy, campaign cleaning, release time and stability commitments.

8.4 Cold chain and logistics

The company describes cold rooms and stability storage at -20°C and 2-8°C but does not publish commercial shipment lanes, qualified packaging systems, excursion limits, data-logger practices, lane validation, customs brokers, freight forwarders or South Africa-specific lead times.[12][18] API

temperature requirements must be product- and stability-data-specific; “cold room available” is not cold-chain qualification.

8.5 Sterile fill, container closure, devices and packaging

If Shilpa proposes an integrated sterile product, Jadcherla creates an acute dependency because of its warning-letter/import-alert history and the historical FDA vial-seal complaint finding.[10][11] The device/cartridge offer is insufficiently located and documented.[16] Require container-closure integrity validation, component vendor qualification, extractables/leachables, silicone/tungsten risk where applicable, media-fill and smoke-study packages, sterilisation/depyrogenation validation, visual-inspection qualification, shipping studies, device master records, assembly/packaging site, human-factors responsibility and current regulator scope.

8.6 Subcontracting

No complete public map of third-party or intra-group subcontracting was found. Unit VII may provide analytical work, while formulation, biologics and device-related operations sit in other legal entities/si

tes.[3] The supplier must declare every laboratory, micronisation/milling step, microbiology/endotoxin test, sterilisation service, reference-standard source, packaging site, storage depot and freight handoff. No undeclared subcontracting should be permitted.

9. South Africa fit and RFI conditions

SPLL is a plausible ****peptide API/drug-substance candidate**** for a South African applicant because desmopressin and octreotide have public CEP evidence and its API plants have recent exposure to FDA, PMDA and ANVISA oversight.[4][5][6][7][8] It is not publicly evidenced as a South African registered peptide supplier, and no SAHPRA product record or label was verified. The South African applicant must independently determine dossier route, API source acceptability, GMP evidence, local licensing, import controls, pharmacopoeial specification and stability-zone requirements.

Progress to confidential RFI only if Shilpa provides:

1. corporate registration documents, current ownership chart and contracting entity;
2. an exact site/building/equipment map for each proposed peptide, including CEP/DMF site linkage;
3. current CEP copies and EDQM status extracts for desmopressin/octreotide; DMF status/access documentation where relevant;
4. FDA Establishment Inspection Reports, Forms 483, responses, inspection classifications and CAPA-effectiveness evidence for API Units I and II;
5. current GMP certificates/inspection scopes from EU authority, PMDA, ANVISA, TGA, MFDS and others claimed, limited to the proposed site/activity;
6. a live FDA warning-letter/import-alert status package for Jadcherla and all 2025 inspection records if any finished-dose work is proposed;
7. representative peptide batch records, process validation, cleaning validation, impurity fate/purge studies, residual-solvent/TFA controls, method validation/transfer, CoAs and stability trends;
8. data-integrity governance, audit-trail review, OOS/OOT/deviation/CAPA metrics and recent product-quality reviews;
9. raw-material supplier/country/dual-source inventory, change-notification and business-continuity mapping;
10. full subcontractor and intra-group service map;
11. cold-chain and South Africa lane qualification; and
12. for sterile/device scope, current line qualification, media fills, smoke studies, CCIT, extractables/leachables, visual inspection, device/packaging responsibility and regulator authorisation.

****Desk decision:**** ****Tier B.**** Invite an RFI for a defined peptide API, prioritising the CEP-backed molecules. Do not prequalify a generic “integrated peptide plus fill-finish/device” platform, do not accept research-only catalogue material, and do not infer any South African approval or customer.

References

[1]: Shilpa Medicare Limited, ****Annual Report 2025–26**** (audited corporate filing), <https://vbshilpa.com/pdf/Annual-report-results-2025-2026.pdf>

[2]: Shilpa Medicare Limited, ****Annual Reports / corporate identity page****, <https://vbshilpa.com/annual-reports.php>

[3]: Shilpa Medicare Limited, ****Locations****, <https://www.vbshilpa.com/locations.php>

[4]: Shilpa Medicare Limited, ****Shilpa Pharma Lifesciences Limited received CEP from EDQM for desmopressin**** (3 October 2024 stock-exchange disclosure), <https://www.vbshilpa.com/pdf/Shilpa%20Pharma%20Lifesciences%20Limited,%20Received%20CEP%20from%20EDQM%20for%20A%20PI,%20Desmopressin.pdf>

[5]: Shilpa Medicare Limited, ****Shilpa Pharma Lifesciences Limited received CEP from EDQM for octreotide**** (18 November 2024 stock-exchange disclosure), <https://www.vbshilpa.com/pdf/Shilpa%20Pharma%20Lifesciences%20Limited,%20Received%20CEP%20from%20EDQM%20for%20A%20PI,%20Octreotide.pdf>

[6]: Shilpa Medicare Limited, ****SPLL Unit I received EIR from USFDA; VAI classification**** (14 May 2025 stock-exchange disclosure), <https://www.vbshilpa.com/pdf/Shilpa%20Pharma%20Lifesciences%20Limited,%20Unit-1,%20Received%20EIR%20from%20USFDA.pdf>

[7]: Shilpa Medicare Limited, ****SPLL Unit II PMDA inspection**** (1 September 2023 stock-exchange disclosure), <https://www.vbshilpa.com/pdf/Shilpa%20Pharma%20Lifesciences%20Ltd,%20Unit%20II,%20Raichur,%20Karnataka%20has%20cleared%20PMDA%20Japan,%20GMP%20inspection.pdf>

[8]: Shilpa Medicare Limited, ****SPLL Unit II ANVISA inspection and GMP certification**** (9 July 2024 stock-exchange disclosure), <https://www.vbshilpa.com/pdf/Shilpa%20Pharma%20Lifesciences%20Limited,%20Unit%20II,%20Raichur,%20Karnataka%20has%20cleared%20the%20ANVISA-Brazil%20GMP%20inspection.pdf>

[9]: Shilpa Medicare Limited, ****AGES EU GMP inspection of Unit IV, Jadcherla**** (16 April 2024 stock-exchange disclosure), <https://www.vbshilpa.com/pdf/Europe%20GMP%20Approval%20of%20Shilpa%20Medicare%20Ltd.%20Unit%20IV,%20Jadcherla,%20Telangana.pdf>

[10]: U.S. Food and Drug Administration, ****Warning Letter 320-21-01, Shilpa Medicare Limited, MARCS-CMS 607877**** (9 October 2020), <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/shilpa-medicare-limited-607877-10092020>

[11]: U.S. Food and Drug Administration, ****Import Alert 66-40: Detention Without Physical Examination of Drugs From Firms Which Have Not Met Drug GMPs**** (current page published 17 August 2026), https://www.accessdata.fda.gov/cms_ia/importalert_189.html

[12]: Shilpa Pharma Lifesciences Limited, ****Capabilities / manufacturing site and peptide purification, lyophilisation and cold-storage claims****, <https://shilpapharma.com/capabilities/>

[13]: Shilpa Pharma Lifesciences Limited, ****Peptide manufacturers / product and regulatory-status**

catalogue**, <https://shilpapharma.com/peptide-manufacturers/>

[14]: Shilpa Pharma Lifesciences Limited, **Peptide CDMO services**, <https://shilpapharma.com/peptide-cdmo-services/>

[15]: Shilpa Medicare Limited, **GMP peptides**, <https://vbshilpa.com/peptides-gmp.php>

[16]: BioPharm International, **Shilpa Medicare launches 'Hybrid CDMO' at DCAT Week 2025** (reports company press-release claims), <https://www.biopharminternational.com/view/shilpa-medicare-launches-hybrid-cdmo-dcat-week-2025>

[17]: Shilpa Medicare Limited, **Non-GMP peptide services**, <https://vbshilpa.com/peptides-non-gmp-services.php>

[18]: Shilpa Pharma Lifesciences Limited, **Analytical services**, <https://shilpapharma.com/analytical-services/>

[19]: European Directorate for the Quality of Medicines & HealthCare, **EDQM databases / Certification database access**, <https://www.edqm.eu/en/databases>

[20]: U.S. Food and Drug Administration, **Inspection Classification Database**, <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/inspection-classification-database>

[21]: South African Health Products Regulatory Authority, **Registered health products database**, <https://www.sahpra.org.za/registered-health-products/>

[22]: Shilpa Medicare Limited, **Press releases and regulatory disclosures index**, <https://www.vbshilpa.com/press-releases.php>

[23]: Shilpa Medicare Limited, **Generic peptide equipment and process claims**, <https://www.vbshilpa.com/peptides-generic.php>

[24]: U.S. Food and Drug Administration, **NDA 215179 Complete Response letters to Shilpa Medicare Limited (pemetrexed injection; non-peptide context)**, https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215179Orig1s000OtherActionLtrs.pdf

****Evidence discipline:**** Company webpages and investor/stock-exchange disclosures are identified as such. Regulator-primary evidence is given greater weight. Claims of market reach, “approval,” scale and capability are not converted into customer, product-approval or site-scope conclusions where the underlying regulator record was not obtained.

07-M07.md

PeptiStar Inc. (Japan) — Therapeutic-Peptide CDMO Desk Dossier

****Dossier ID:** 07-M07**

****Decision date:** 20 August 2026**

****Purpose:** Preliminary prequalification for regulated therapeutic supply to a South African business**

****Preliminary desk tier:** **B — credible manufacturer, but material public-evidence gaps****

1. Decision summary

PeptiStar Inc. is a Japanese private company established on 1 September 2017, corporate number ****5120901038915****, with its registered head office and disclosed manufacturing campus at ****2-5-1 Mishima, Settsu, Osaka 566-0022, Japan****. Japan's Ministry of Economy, Trade and Industry (METI) gBizINFO record corroborates the entity, corporate number and address; PeptiStar's company outline identifies Shinichiro Fujiie as representative director and describes the business as development, manufacture and sale of special-peptide and oligonucleotide active pharmaceutical ingredients (APIs).[1][2]

PeptiStar originated as a joint venture among PeptiDream Inc., Shionogi & Co., Ltd. and Sekisui Chemical Co., Ltd.[3] No single current parent is publicly identified. PeptiDream's audited FY2025 disclosure says that it held ****less than 20%**** at reporting date and describes PeptiStar as an equity holding rather than a consolidated subsidiary; PeptiStar also disclosed 2024–2025 new-share subscriptions by Sumitomo Chemical, Toagosei, Shionogi, Sekisui Chemical, Shimadzu and the Development Bank of Japan, plus PeptiDream's acquisition of shares from existing holders including INCJ.[4][5][6][7] A complete current cap table, voting rights and ultimate-control analysis are not public and must be obtained in the RFI.

The disclosed scope is ****peptide API/drug-substance development, analytical/CMC support, clinical API and commercial API manufacture****. PeptiStar shows substantial self-reported synthesis, preparative purification and bulk lyophilization capability at one Settsu campus, including 1–300 L filter reactors, 1–30 L microwave SPPS equipment, columns up to 450 mm diameter and 3–500 L lyophilizers.[8][9] It reports a maximum actual yield above 1,000 g per batch in Manufacturing Building 1 and states that its larger process can support several hundred kilograms per year.[8][10] These are ****company marketing/technical claims****, not independently verified installed-capacity or validated-state evidence.

No public evidence reviewed establishes that PeptiStar performs ****sterile finished-dose manufacture, aseptic fill-finish, cartridge/syringe/vial filling, device assembly, labelling or final packaging****. Class 100,000 areas described by the company apply to purification and bulk API lyophilization and must not be treated as evidence of an aseptic filling suite.[8][10][11] Research-grade and sample services are offered alongside GMP work; any research-only batch must be excluded from therapeutic supply unless it is transferred, validated, manufactured, tested and released under the applicable GMP and dossier controls.[12]

The principal desk weakness is the absence of publicly located, PeptiStar-specific regulator evidence tying the Settsu campus to an approved therapeutic product or an inspection outcome. Targeted exact-name/address searches through the public sources specified for this review did not locate a PeptiStar-specific FDA inspection classification or EIR reference, FDA warning letter/import alert, EudraGMDP GMP certificate/non-compliance report, EDQM CEP, PMDA/MHLW inspection certificate, or public NMPA, MFDS, WHO, TGA, Health Canada or SAHPRA record as of the decision date. This means ****not publicly evidenced****, not that no confidential filing, inspection or compliant operation exists. FDA explains that its classification database concerns inspections related to currently marketed FDA-regulated products; EMA describes EudraGMDP as the public repository for manufacturing authorisations, GMP/GDP certificates, API registrations and

non-compliance reports.[13][14]

Accordingly, ****Tier B**** is ap

appropriate: PeptiStar is credible enough for a formal RFI and on-site/remote quality audit, but the public record does not support Tier A regulated-market confidence. It must not be described as fully compliant, SAHPRA-approved or safe.

2. Legal entity, ownership and governance

Item	Finding	Evidence status
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| Legal name | **PeptiStar Inc.**; Japanese name **■■■■■■■■■■■** | Corroborated by company
and METI government record.[1][2] |

| Corporate number | **5120901038915** | METI gBizINFO/National Tax Agency-linked record.[2] |

Incorporation | **1 September 2017** | Company and government record.[1][2] |

| Registered/head-office address | **2-5-1 Mishima, Settsu, Osaka 566-0022, Japan** | Company
and government record.[1][2] |

| Representative | ****Shinichiro Fujiie****, Representative Director / President & CEO | Company and government record.[1][2] |

| Founders | PeptiDream, Shionogi and Sekisui Chemical | PeptiDream corporate history and audited disclosure.[3][4] |

| Current parent | **No parent publicly identified.** | PeptiDream held under 20% at FY2025 reporting; multiple investors subscribed in 2024–2025.[4][5][6][7] |

| Current ownership | Incomplete public picture; includes or has included the founders and later investors named above. Exact percentages, voting rights, preferred rights and ultimate control are undisclosed. | RFI gap. |

The METI page reports one social-insurance establishment at the same Settsu address and 67 insured persons, but this is not a pharmaceutical-manufacturing licence and should not be used as GMP evidence.[2] Likewise, the Japanese government procurement qualification shown for manufacture/sale of pharmaceuticals and medical supplies is an eligibility classification, not a medicinal-product approval or GMP certificate.[2]

3. Sites and roles

All disclosed operational buildings are on ****one Settsu campus****. They should be treated as separate functional buildings within a potentially single licensed site until PeptiStar supplies the Japanese manufacturing-establishment licence and its building/room annexes.

| Site/building | Publicly described role | Therapeutic-supply interpretation |

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| ****HQ/R&D Building****, 2-5-1 Mishima, Settsu | Administration; process and analytical method development; samples; medium-scale synthesis; small-scale GMP manufacture; GMP testing; application and audit support. Equipment pictured/listed includes microwave peptide synthesis, HPLC, GC, quadrupole MS and other MS.[15] | Mixed R&D and claimed GMP operations. Segregation, validated systems, material/personnel flows, status labelling and controls preventing research/GMP cross-use require audit. |

| ****Manufacturing Building 1****, same campus | Two claimed GMP lines for peptide and oligonucleotide APIs; synthesis separated from purification; purification in Class 100,000; up to four products simultaneously; maximum actual yield stated as over 1,000 g/batch.[10] | Bulk API/drug substance only on public evidence. Campaign controls and multiproduct cleaning validation are critical. |

| **Manufacturing Building 2**, same campus | Large peptide/oligonucleotide API facility; one claimed GMP line with synthesis and purification separated; Class 100,000 purification; up to two products simultaneously; 300 L filter reactor and lyophilizer shown.[11] | Bulk API/drug substance only. No sterile filling is evidenced. |

The company states the three-building campus has about **10,500 m²** total floor space, with the manufacturing buildings covering investigational through commercial production.[16] No additional PeptiStar-owned peptide manufacturing site or foreign subsidiary was identified in the reviewed primary disclosures. Sumitomo Chemical's Oita Works is a separate Sumitomo site in a disclosed dual-site gRNA arrangement and must **not** be attributed to PeptiStar's peptide manufacturing network.[17]

Single-site concentration

PeptiStar's disclosed peptide development, sc

ale-up, manufacturing, QC and release-support functions are concentrated at the Settsu address. Multiple buildings offer some internal redundancy, but they remain exposed to the same local utilities, natural-hazard, workforce, logistics and site-governance risks. Business-continuity plans, alternate QC/release arrangements, validated backup utilities, disaster recovery and second-source strategy are not publicly disclosed.

4. Service-scope boundary

| Category | Finding at decision date |

|---|---|

| **Peptide API / drug substance** | **Yes, claimed core capability.** Process development, analytical development, clinical API, commercial API, purification and bulk lyophilization are explicitly offered.[8][9][12] |

| **Sterile finished dose** | **Not publicly evidenced.** No formulation suite, sterile dosage-form release, terminal sterilisation or finished-product manufacturing service was identified. |

| **Aseptic fill-finish** | **Not publicly evidenced.** Class 100,000 purification/bulk drying does not demonstrate Grade A/B aseptic filling. |

| **Device / combination-product assembly** | **Not publicly evidenced.** PDC means peptide-drug conjugate, not a delivery device.[5] |

| **Primary/secondary packaging, labelling, serialisation** | **Not publicly evidenced.** Bulk API packaging configuration is also undisclosed. |

| **Analytical/QC and stability** | **Yes, self-described**, including method development/validation, specifications, quality testing and ICH-aligned stability studies.[18] |

| **Regulatory/CMC support** | **Yes, self-described**, including application-document preparation, agency-question support and English documentation for EMA/FDA review.[19] This does not evidence an accepted dossier. |

| **Research/sample services** | **Yes.** The company offers few-milligram samples and research-grade through commercial work.[12][20] Research-only material is outside therapeutic consideration. |

| **Other modalities** | Oligonucleotide API and PDC services are disclosed but are outside this peptide-focused decision except where they create multiproduct or shared-system risk.[5][12] |

Any South African supply model would therefore require a separately qualified **sterile finished-dose/fill-finish and packaging partner**, unless PeptiStar provides non-public evidence of qualified subcontractors and the applicant's quality/technical agreements assign full responsibilities.

5. Peptide manufacturing capability

5.1 Synthesis and scale

PeptiStar describes conventional Fmoc solid-phase peptide synthesis (SPPS), microwave-assisted SPPS and liquid-phase peptide synthesis (LPPS). Its public equipment list includes 1, 10, 30 and 300 L filter reactors. It states that a 300 L reactor can support commercial requirements of several hundred kilograms per year, while its homepage gives a broader few-milligram to

several-dozen-kilogram GMP range.[8][20] These different measures appear to refer respectively to annual throughput and campaign/batch scale, but batch definitions, solids loading, utilisation, sequence-dependent yield and demonstrated commercial campaign data are not public.

Microwave SPPS equipment is reported at 1, 10 and 30 L, jointly developed with Microwave Chemical and PeptiDream; PeptiStar says it spans research to commercial use and reduces reaction time and amino-acid usage.[21] It also claims capability for peptides longer than 100 residues, cyclic, branched and cyclic-dimer peptides and conjugates.[8] No public batch record, process-performance qualification (PPQ), annual product review or approved-product linkage independently verifies these claims.

The company's STag-PS LPPS platform is jointly developed with Sekisui Medical and is described as enabling in-process HPLC monitoring, one-pot coupling and reduced amino-acid/solvent use.[22] SYNC SOL, licensed from technology developed by PeptiDream and Nissan Chemical, uses unprotected amino-acid coupling and a silyl organic tag; the company states that i

t can reduce coupling-agent use and control epimerisation.[23] Public information does not show which projects have adopted these platforms under GMP or whether any regulatory authority has reviewed them.

5.2 Purification, concentration and lyophilization

The company reports preparative columns from 10 to 450 mm diameter and processing from purification onward in Class 100,000 rooms.[8] It offers reverse-phase and complementary HILIC/SFC-mode separations, noting that an impurity can co-elute with the main peptide on an ODS reverse-phase method and that the Daicel DCPak P platform provides orthogonal resolution.[24] Continuous chromatography developed with YMC is described as recyclable-fraction purification, 24/7 automated operation, reduced fraction loss and GMP-capable operation.[25] These statements plausibly address difficult closely related impurities, but product-specific impurity fate/purge data, carryover limits, resin/column lifetime and validated automation controls are not public.

Lyophilizers of 3, 30, 50 and 500 L are listed for bulk API drying in Class 100,000 conditions.[8] The mixer-type RHEOFREED platform, jointly developed with Kobelco Eco-Solutions, is claimed to shorten drying time by more than 80% and improve powder homogeneity.[26] The public description does not establish validated residual-solvent/moisture endpoints, sterile operation, container closure integrity, or finished-dose lyophilisation.

Asahi Kasei reported on 25 June 2026 that its 100 L forward-osmosis/membrane-distillation concentration system had been installed at PeptiStar for trial API production and was operating at manufacturing scale **as part of evaluation toward GMP production**.[27] Therefore this novel FO–MD system must be treated as **not yet publicly evidenced for GMP production** at the decision date. It also creates proprietary membrane, equipment-support and change-control dependencies on Asahi Kasei.

6. Analytical, release and stability evidence

PeptiStar's own analytical page lists the following:

- * identity by mass spectrometry and amino-acid analysis;
- * related substances and assay/content by HPLC/UHPLC;
- * counter-ion by HPLC/ion chromatography;
- * water by Karl Fischer and thermal analysis;
- * residual solvents by GC/GC-MS;
- * endotoxin, microbial limits and elemental impurities;
- * peptide-focused LC-MS/MS, UV and IR structural analysis;
- * handling of moisture-sensitive samples in a controlled-temperature/humidity weighing room and

highly toxic/genotoxic impurities in a balance enclosure; and

* long-term, accelerated, stress and photostability studies described as conforming to ICH guidelines.[18]

This is a useful claimed platform for API specifications, release and stability, but the site does not publish product-specific validated methods, acceptance criteria, reference-standard qualification, chromatographic integration governance, audit-trail review, stability protocols, sample accountability, microbiology laboratory scope, method-transfer performance or proficiency testing. It is also unclear which tests are performed in-house versus outsourced.

PeptiStar says the HQ/R&D building performs GMP testing and lists HPLC, GC and MS equipment.[15] Its quality page describes management review, meetings of GMP-responsible persons, self-inspection, product-quality review and supplier qualification; it further claims company-wide GxP data-integrity compliance with ALCOA and CCEA principles and training-controlled system access/work.[28] These are ****manufacturer assertions****, not regulator findings or a published independent audit.

Impurity-control assessment

The public technical package shows awareness of peptide-specific risks: deletion/addition sequences and closely related impurities (orthogonal RP-HPLC/HILIC/SFC), epimerisation (SYNCSOL claim), residual solvent, counter-ion, water, elemental impurity, endotoxin, bioburden/microbial limits and potentially genotoxic impurities.[18][23][24] However, no ev

idence was located for a formal product-specific impurity-control strategy covering starting-material impurity carryover, amino-acid stereochemical purity, nitrosamines, mutagenic coupling-reagent by-products, aggregation, oxidation/deamidation, disulfide scrambling, column leachables, membrane extractables/leachables or cross-contamination limits. These must be requested by molecule and route.

7. Pharmaceutical quality system and data governance

PeptiStar publicly claims a company-wide cGXP quality system, management involvement, self-inspection, product-quality review, lifecycle management, supplier qualification and ALCOA/CCEA-based data-integrity controls.[28] It does not publicly describe:

- * deviation classification and investigation effectiveness;
- * ****OOS**** laboratory/manufacturing investigation procedure or invalidation governance;
- * ****OOT**** trend detection and escalation;
- * CAPA due-date control, effectiveness checks and recurrence metrics;
- * change-control governance and regulatory-impact assessment;
- * electronic-system inventory, validation, access recertification and audit-trail review frequency;
- * raw-data backup, disaster recovery and cybersecurity controls;
- * quality-metric trends, recalls, rejected batches or critical deviations;
- * PPQ/continued process verification status;
- * cleaning validation and health-based exposure limits for multiproduct peptide/oligonucleotide lines; or
- * quality-unit independence, authorised batch-release roles and disposition authority.

These are ****unknowns****, not adverse findings. An RFI should obtain SOP indices, recent metrics and anonymised representative investigations before audit.

8. Regulatory and market evidence

8.1 Publicly evidenced regulatory position

The company says it performs small- and commercial-scale GMP manufacture and assists with

EMA/FDA application documentation.[10][11][15][19] No primary regulator document reviewed corroborates a PeptiStar inspection, acceptable classification or product approval. Public exact-name/address checks found no PeptiStar-specific evidence in the searched FDA inspection/classification, warning-letter/import-alert, EMA/EudraGMDP, EDQM CEP, PMDA/MHLW, NMPA, MFDS, WHO, TGA, Health Canada or SAHPRA sources as of 20 August 2026. Absence from a public search may reflect confidentiality, database scope, name/address variation, sponsor-held filings or no relevant event; it must not be interpreted as either compliance or non-compliance.

No public FDA **EIR** reference, Form FDA 483, final inspection classification, warning letter or import alert was identified. No public EU GMP certificate, EU non-compliance statement, active-substance registration or CEP naming PeptiStar was identified. No public Japanese inspection outcome or product-specific GMP conformity record naming the site was identified. No approval label, assessment report, dossier number, DMF number/letter of authorisation or product name was identified that would link PeptiStar to a marketed peptide medicine.

8.2 Markets/countries served

PeptiStar states generally that it has domestic and overseas clients and provides services to clients in Japan and internationally.[6][29] This does **not** identify a regulated market, customer, product approval or commercial supply country. Its CMC page mentions document support for EMA and FDA screening, but not an accepted submission.[19]

Accordingly, **no specific regulated therapeutic market or country served is publicly evidenced for PeptiStar peptide API supply**. The United States, European Union and Japan should not be listed as supplied markets without a customer-authorised DMF/CEP/ASMF reference, regulator inspection evidence, approved dossier/label, or equivalent product-specific documentation. No customer is inferred in this dossier.

9. Adverse regulatory history search

No PeptiStar-specific warning letter, import alert, regulator non-compliance statement, recall, consent decree, administrative sanction

or public enforcement action was located in the targeted sources reviewed through the decision date. This is a **negative desk-search result with limitations**, not proof of a clean history. Searches should be repeated during live RFI review using Japanese names, prior names, site identifiers, FEI/DUNS/UEI and sponsor-authorised records.

PeptiStar should provide a signed global regulatory-history declaration covering inspections, observations, commitments, enforcement, recalls, market actions, serious quality defects, data-integrity allegations, licence restrictions and pending litigation since incorporation, plus copies of all inspection reports and CAPA close-outs that it is permitted to share.

10. Supply-chain and operational dependencies

| Dependency | Evidence and risk interpretation |

|---|---|

| Protected amino acids, specialty/non-natural amino acids, coupling reagents and SPPS resins | PeptiStar itself identifies excessive protected-amino-acid and coupling-reagent use as a conventional SPPS cost issue.[9] Suppliers, countries of origin, safety stocks, retest periods and second sources are undisclosed. Resin is intrinsic to SPPS but no resin brand/supplier is publicly named. |

| Solvents and counter-ions | HPLC purification and FO–MD descriptions expressly involve water and volatile organic solvents including acetonitrile; residual-solvent and counter-ion testing is offered.[18][27] Solvent recovery, shortage contingency and hazardous-logistics arrangements are unknown. |

| Proprietary/licensed process technologies | Microwave SPPS depends on Microwave Chemical/PeptiDream collaboration; STag-PS on Sekisui Medical; SYNC SOL on PeptiDream/Nissan Chemical; chromatography on YMC; orthogonal columns on Daicel; mixer lyophilisation on Kobelco Eco-Solutions; FO–MD on Asahi Kasei.[21]–[27] Licence continuity, spare parts, consumables and change-notification rights require diligence. |

| Single campus | All disclosed peptide operations are in Settsu. Multiple buildings do not eliminate common-site disaster, utility or access risk.[16] |

| Multiproduct operations | Buildings manufacture peptides and oligonucleotides, with several products potentially simultaneous.[10][11] Cleaning validation, segregation, campaign control and cross-contamination risk require audit. |

| Cold chain and shipping | Storage temperatures, cold rooms/freezers, mapping, alarm response, shipping lanes, data loggers, excursions, dangerous-goods classification and South Africa lane qualification are not publicly disclosed. Product-specific stability data must set transport conditions. |

| Sterile fill and packaging | No in-house capability is publicly evidenced. A separate qualified contractor and quality agreement would be necessary for a sterile finished dose. Subcontractor identity and oversight are unknown. |

| QC subcontracting | The public test menu does not say which assays are outsourced. Technical/quality agreements, audit status, sample chain of custody and data review are unknown. |

| Novel FO–MD system | Publicly still under evaluation toward GMP as of June 2026; do not place it in a registered process without completed qualification, validation and dossier/change-control support.[27] |

| Geopolitical/logistics | Japan is generally stable, but imported amino acids/resins/solvents or globally sole-sourced specialty reagents may create exposure. Actual source countries are undisclosed and must not be inferred. Ocean/air freight to South Africa creates lead-time and temperature-excursion risk. |

11. South Africa fit and required RFI gates

PeptiStar is a plausible ****non-sterile peptide API/drug-substance**** candidate for a South African sponsor, especially for complex synthetic peptides requiring SPPS/LPPS, preparative chromatography and bulk lyophilisation. The public platform appears technically relevant, and the entity/site identity is clear. The present evidence is nevertheless insufficient for selection, dossier reliance or commercial commitment.

Progress should be limited to a confidential RFI and risk-based audit, conditional on receiving:

1. the current Japanese manufacturing-establishment licence, scope/annexes and latest PMDA/MHLW/prefectural GMP conformity evidence for the exact Settsu buildings and operations;
2. all regulator inspections since start-up, including dates, scope, classifications, observations, responses, CAPA effectiveness and any EIR/GMP certificates;
3. product- and market-specific evidence: customer-authorised DMF/ASMF/CEP/CTD references, accepted letters of access, approved-product linkage and commercial-batch history, with customer names redacted if necessary;
4. current ownership/cap table, ultimate beneficial ownership, board control and licence/IP continuity;
5. site master file, organisation chart, quality manual, validation master plan and exact authorised batch-release chain;
6. product-specific process description, demonstrated scale, PPQ/continued-verification status, impurity fate/purge, genotoxic/nitrosamine risk assessments and comparability strategy;
7. cleaning validation and health-based exposure limits for shared peptide/oligonucleotide equipment, plus campaign and simultaneous-product controls;
8. data-integrity governance, computer-system validation, audit-trail review, backup/restore testing

13. identification and separate qualification of any sterile fill-finish, device or packaging partner.

No therapeutic purchase should rely on research-grade material, a certificate of analysis alone, marketing statements, or the presence of a Class 100,000 room. SAHPRA filing strategy, GMP evidence acceptability and dossier right of reference must be confirmed by the South African applicant and regulatory counsel.

12. Preliminary tier rationale

****Tier B — credible but material gaps.**** Positive factors are a clearly resolved Japanese legal entity; an established, physically described single campus; founding and investment by major Japanese life-science/industrial companies; plausible peptide API equipment; and a broad self-described analytical and quality platform. Tier A is not justified because the review found no public regulator inspection classification/certificate, accepted DMF/CEP/ASMF, named approved therapeutic product, product label or specific regulated-market supply record linking PeptiStar's Settsu site to therapeutic peptide API. Sterile finish, fill-finish, device/packaging, subcontracting, cold chain, OOS/OOT/CAPA effectiveness and detailed data-integrity evidence remain material gaps.

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References

- [1]: PeptiStar Inc., "Company Outline" (Japanese) <https://peptistar.com/company/outline/>
- [2]: Ministry of Economy, Trade and Industry, gBizINFO, "[REDACTED]," <https://info.gbiz.go.jp/hojin/ichiran?hojinBango=5120901038915>
- [3]: PeptiDream Inc., "Company History," <https://www.peptidream.com/en/company/history/>
- [4]: PeptiDream Inc., "Consolidated Financial Results for the Fiscal Year Ended December 31, 2025 [IFRS]," pp. 22–23, <https://contents.xj-storage.jp/xcontents/45870/0f4e7954/5cfb/4ea6/97e3/9737ca27ce47/140120260216562906.pdf>
- [5]: PeptiStar Inc., "Notice of fundraising," 11 November 2024, <https://peptistar.com/en/news/20241111-2/>
- [6]: PeptiStar Inc., "Notice of fundraising," 26 March 2025, <https://peptistar.com/en/news/20250326-2/>
- [7]: PeptiStar Inc., "Notice of fundraising," 17 April 2025, <https://peptistar.com/en/news/20250417-2/>
- [8]: PeptiStar Inc., "Peptide API Development and Manufacturing," <https://peptistar.com/en/service/peptide/>
- [9]: PeptiStar Inc., "Technology," <https://peptistar.com/en/technology/>
- [10]: PeptiStar Inc., "Manufacturing Building 1," <https://peptistar.com/en/facility/manufacturing1/>
- [11]: PeptiStar Inc., "Manufacturing Building 2," <https://peptistar.com/en/facility/manufacturing2/>
- [12]: PeptiStar Inc., "Service," <https://peptistar.com/en/service/>
- [13]: U.S. Food and Drug Administration, "Inspection | Enforcement Resources," content current 12 December 2025, <https://www.fda.gov/drugs/pharmaceutical-quality-resources/inspection-enforcement-resources>
- [14]: European Medicines Agency, "EudraGMDP database," <https://www.ema.europa.eu/en/human-regulatory-overview/research-development/compliance-research-development/good-manufacturing-practices>

g-practice/eudragmdp-database

- [15]: PeptiStar Inc., "HQ R&D Building," <https://peptistar.com/en/facility/headquarters/>
- [16]: PeptiStar Inc., "Facility," <https://peptistar.com/en/facility/>
- [17]: Sumitomo Chemical Co., Ltd., "Sumitomo Chemical Subscribes for Shares of PeptiStar in Private Placement," 11 November 2024, <https://www.sumitomo-chem.co.jp/english/news/detail/20241111e.html>
- [18]: PeptiStar Inc., "Analysis," <https://peptistar.com/en/service/analysis/>
- [19]: PeptiStar Inc., "Support for Clinical Trial and Application for Approval," <https://peptistar.com/en/service/cmc/>
- [20]: PeptiStar Inc., homepage, <https://peptistar.com/en/>
- [21]: PeptiStar Inc., "MW SPPS Method," <https://peptistar.com/en/technology/microwave/>
- [22]: PeptiStar Inc., "STag-PS Method," <https://peptistar.com/en/technology/stagps/>
- [23]: PeptiStar Inc., "SYNCSOL," <https://peptistar.com/en/technology/syncsol/>
- [24]: PeptiStar Inc., "HILIC," <https://peptistar.com/en/technology/hilic/>
- [25]: PeptiStar Inc., "Continuous Chromatography Purification Method," <https://peptistar.com/en/technology/mcsgp/>
- [26]: PeptiStar Inc., "Mixer-type Lyophilization," <https://peptistar.com/en/technology/lyophilization/>
- [27]: Asahi Kasei Corp., "Asahi Kasei's novel forward osmosis–membrane distillation system installed at Peptistar's API manufacturing facility," 25 June 2026, <https://www.asahi-kasei.com/news/2026/e260625>
- [28]: PeptiStar Inc., "Quality," <https://peptistar.com/en/quality/>
- [29]: PeptiStar Inc., "Message," <https://peptistar.com/en/company/message/>

****Cut-off statement:**** Sources were assessed for information available through ****20 August 2026****. Public-record absence statements are bounded by database coverage, indexing, name/address variants, sponsor confidentiality and access limitations.

08-M08.md

08-M08 — Ajinomoto Bio-Pharma Services (Japan)

****Decision date:**** 20 August 2026

****Purpose:**** Preliminary desk prequalification for regulated therapeutic-peptide supply to a South African business

****Scope:**** Japan operations and material current/former affiliates or subcontracting arrangements needed to understand peptide drug-substance and downstream supply

****Preliminary desk tier:**** ****B** — credible, but material gaps

Executive conclusion

Ajinomoto Bio-Pharma Services is a ****service brand and operating organization, not a single global legal entity****. In Japan, the principal contracting/manufacturing entities identified from primary corporate sources are ****Ajinomoto Co., Inc.**** (Tokyo headquarters and the Tokai/AJIPHASE operation) and its wholly owned subsidiary ****GeneDesign, Inc., doing business as Ajinomoto Bio-Pharma Services**** (Osaka). Ajinomoto Co., Inc. is the listed parent (Tokyo Stock Exchange code 2802). The company stated in 2018 that the unified brand would continue to operate through its existing legal structures.[1] Current corporate pages still identify GeneDesign, Inc. as the Osaka legal company and as doing business under the Ajinomoto Bio-Pharma Services name.[2][3]

For therapeutic peptides, the clearest relevant manufacturing proposition is ****AJIPHASE hybrid liquid/solid-phase chemical synthesis at Tokai, Japan****. Corporate materials claim high-purity oligonucleotide and peptide manufacture up to ****12,000 L**** at Tokai, with AJIPHASE peptide output in ****kilogram quantities under cGMP**** and broader platform scale up to 200 kg.[4][5][6] These are ****manufacturer claims****, not independently verified batch records or regulator certificates. The Osaka facility has cGMP oligonucleotide API infrastructure and lists contracted peptide synthesis, but the public scale tables, purification and lyophilization details are explicitly for oligonucleotides rather than peptides; Osaka should therefore not be treated as a proven commercial peptide API site without an RFI response and site-specific evidence.[2][7][8]

Public regulated-market evidence is meaningful at group/platform level but ****not peptide-specific enough for Tier A****. Ajinomoto announced that an undisclosed commercial drug made using AJIPHASE obtained US FDA approval in 2020, but the announcement expressly describes expansion of AJIPHASE into oligonucleotides and does not identify a peptide product, sponsor, manufacturing site, application number or FDA assessment record.[9] No peptide-specific US approval, EU GMP certificate, CEP, Japanese GMP compliance result, PMDA product review, TGA/Health Canada record, WHO record, NMPA/MFDS record or South African registration naming Tokai or GeneDesign was located in the public sources reviewed. This does not prove absence; it creates an RFI-critical evidence gap.

The group has current adverse FDA inspection evidence at Ajinomoto Co.'s ****Kawasaki Plant****, an API facility separate from the Tokai AJIPHASE and Osaka GeneDesign sites. FDA's February 2025 Form 483 identified deficient deviation/OOS root-cause and preventive-action investigations, uncontrolled laboratory sample-preparation worksheets, and inadequate evidence that production-tank construction material was non-additive to APIs.[10] A secondary compilation of FDA data reports the inspection as ****VAI****, following NAI outcomes in 2015, 2017 and 2019, but the compilation's facility presentation is imperfect and the final classification/EIR should be obtained directly for due diligence.[11] The findings are relevant to corporate quality-governance questioning but must ****not**** be represented as observations at Tokai or Osaka.

The current Japanese business does ****not**** provide an evidenced in-group sterile drug-product or device/package solution. Ajinomoto sold all shares in Ajinomoto Althea, Inc., its US sterile fill-finish subsidiary, to PCI Pharma Services, completing the transaction on 1 May 2025. Ajinomoto stated that the remaining Bio-Pharma S

ervices business would focus mainly on drug-substance manufacturing.[12] Any sterile fill, finished dose, lyophilized vial, prefilled syringe, device assembly or commercial packaging must therefore be separately qualified and contractually mapped; PCI must not be treated as an Ajinomoto affiliate after completion.

****Desk decision:**** advance to a tightly scoped RFI and quality/technical audit preparation, but do not approve for therapeutic supply on public evidence alone. Tier ****B**** is appropriate because a substantial Japanese parent, identified legal entities, purpose-built sites, cGMP claims, a US-approved AJIPHASE oligonucleotide precedent and public FDA inspection history establish credibility, while peptide-specific regulatory status, site inspection records, commercial peptide references, quality-system evidence and end-to-end sterile/downstream arrangements remain materially incomplete.

1. Identity, ownership and legal-entity resolution

Item	Resolved position	Evidence and qualification
Service brand	**Ajinomoto Bio-Pharma Services**	The 2018 integration announcement says the name unifies multiple businesses but that the companies continued in their existing legal structures.[1] Contracts, quality agreements, purchase orders and regulatory documents must name the performing legal entity, not only the brand.
Ultimate parent	**Ajinomoto Co., Inc.** , Japan; TSE 2802	Ajinomoto's announcement describes Ajinomoto Bio-Pharma Services as wholly owned by Ajinomoto Co., Inc.[1]
Japan AJIPHASE/Tokai operator	**Ajinomoto Co., Inc.**	— confirmation required in the RFI for the exact licensed establishment name The official Japan page identifies Tokai at 1730 Hinaga-cho and the Ajinomoto materials describe AJIPHASE as developed and operated by Ajinomoto Co.; no separate Tokai peptide subsidiary is identified.[4][5] Obtain Japanese manufacturing-business licence, manufacturing-site registration and contracting-entity documents.
Osaka operator	**GeneDesign, Inc.** , doing business as Ajinomoto Bio-Pharma Services	Current company information provides the legal name, DBA and Osaka address; Ajinomoto's global network also labels the location "Ajinomoto Bio-Pharma Services Osaka (GeneDesign, Inc.)."[2][3]
Osaka ownership	**Wholly owned Ajinomoto company**	Ajinomoto's 2019 integration announcement expressly says GeneDesign was wholly owned.[8] The 2016 acquisition structure was initially 95% Ajinomoto Co. and 5% S.A. Ajinomoto OmniChem; later primary material describes it as wholly owned.[13][8] Obtain a current registry extract and authorised-signatory list.
Former US fill-finish entity	**Ajinomoto Althea, Inc.** , formerly d/b/a Ajinomoto Bio-Pharma Services US; sold to PCI Entire Ajinomoto equity stake transferred; completion 1 May 2025.[12]	It is no longer a current affiliate and cannot be included in an Ajinomoto supply proposal without a separately disclosed third-party contract.
Fermentation partner	**Olon S.p.A.** , Italy, independent strategic partner	The October 2025 partnership joins Ajinomoto's CORYNEX development platform to Olon's large-scale GMP microbial fermentation.[14] Olon is a supplier/subcontracting dependency, not an Ajinomoto legal entity.

2. Relevant sites and role boundaries

Site	Legal entity / status	Verified or claimed role	Therapeutic-supply classification
Tokyo — 15-1 Kyobashi 1-chome, Chuo-ku, Tokyo 104-8315	Ajinomoto Co., Inc.	Headquarters for the Pharmaceutical Custom Manufacturing Department.[4] Administrative/business-development site; **not evidenced as peptide manufacturing, sterile fill, testing or packaging** .	
Tokai — 1730 Hinaga-cho, Yokkaichi, Mie 510-0885	Presented as Ajinomoto Co. operation; exact licensed name to confirm Corporate site says high-purity oligonucleotide and peptide large-scale manufacture using AJIPHASE, up to 12,000 L.[4]	AJIPHASE materials claim c	

GMP kg peptide output and broader 200 kg platform capability.[5][6] | **Primary candidate peptide drug-substance/API site.** No public site-specific regulator certificate or identified approved peptide was located. |

| **Osaka — 7-7-29 Saitoasagi, Ibaraki, Osaka 567-0085** | GeneDesign, Inc., d/b/a Ajinomoto Bio-Pharma Services | Head office/laboratory; manufacturer says campus has manufacturing and research centres. GeneDesign lists contracted peptide synthesis, but published facility evidence is strongest for solid-phase oligonucleotide API (Class 100,000 cGMP area; µg–10 kg).[2][4][8] |

Development/research and possible peptide synthesis; commercial therapeutic-peptide status unverified. Treat research/non-GMP output as excluded unless a project is demonstrably transferred into a qualified GMP train. |

| **Kawasaki — 1-1 Suzuki-cho, Kawasaki-ku, Kawasaki, Kanagawa 210-8680** | Ajinomoto Co., Inc. | Japanese R&D base is claimed for CORYNEX strain, fermentation/purification-process and analytical-assay development; GMP manufacture is assigned to partner CMOs worldwide.[15] FDA inspected the address as an API manufacturer in 2025.[10] | **R&D/process-development relevance, not evidenced as the proposed peptide GMP production site.** Do not transfer Kawasaki inspection status to Tokai/Osaka. |

| **Olon sites (not specified in the partnership announcement)** | Olon S.p.A., third party | Announced large-scale CORYNEX peptide/protein GMP fermentation using >100,000 L capability; Olon's stated fermenter range is 1–255 m³. [14] | **Potential outsourced peptide/protein drug-substance manufacture.** Exact site, licence, audit status, responsibilities and regulatory filing must be disclosed per project. |

| **Former San Diego fill-finish — 6175 Lusk Blvd./11040 Roselle St., San Diego, California** | Ajinomoto Althea, Inc.; divested to PCI in May 2025 | Former aseptic drug-product fill/finish/formulation operation; FDA inspected in January 2023 and issued a Form 483.[12][16] | **Not a current Ajinomoto site.** Exclude from an Ajinomoto supply-chain claim unless separately contracted with and qualified as PCI. |

3. Capability assessment by supply-chain stage

3.1 Peptide API/drug substance

The manufacturer describes AJIPHASE as a hybrid of traditional solid-phase and solution-phase synthesis that uses a soluble anchor rather than polymer resin. The anchor-bound intermediate precipitates in polar solvent and is filtered from excess reagents and by-products in one reactor.[5] The company claims lower solvent and reagent use, purity profiles comparable to traditional SPPS, commercial cGMP oligonucleotide manufacture up to 200 kg, and **high-purity peptide manufacture in kilogram quantities under cGMP**.[5][6] Tokai is stated to operate up to 12,000 L.[4]

These statements support a plausible **synthetic peptide drug-substance** capability, but important technical distinctions remain unresolved. The public 200 kg figure is stated for the overall AJIPHASE platform and prominently for oligonucleotides; it should not be treated as a verified 200 kg peptide batch size. The phrase “no isolation required” for peptides appears to concern synthesis/intermediate handling and is not proof that final drug substance needs no purification.[5] Public sources do not provide peptide-specific maximum chain length, modified/non-natural amino-acid scope, cyclisation/disulfide capability, conjugation/lipidation, cleavage chemistry, batch yields, preparative chromatographic load, final salt exchange, milling, sterile filtration, bulk lyophilisation or commercial campaign history.

CORYNEX is a separate **microbial expression** route for long peptides/proteins using *Corynebacterium glutamicum*. Marketing materials claim secretory expression, low impurities, no endotoxin from the host, examples including peptide and GLP-1-related titres around 3 g/L, and in-house Japanese process/assay development.[15] GMP production is not represented as o

ccurring at the Japanese R&D base; it is performed with partner CMOs, and the Olon collaboration expressly targets large-scale drug substance including GLP-1 analogues.[14][15] Treat “endotoxin free expression system” as a platform claim, not proof that a released bulk is free of bacterial contaminants or that endotoxin testing is unnecessary.

3.2 Sterile finished dose and fill-finish

No current Japanese sterile fill-finish capability was evidenced. The only clearly identified group aseptic drug-product operation, Ajinomoto Althea in the United States, was sold to PCI in May 2025. Ajinomoto's transaction notice says the remaining business will focus mainly on drug substance.[12] Consequently, vial or prefilled-syringe formulation, compounding, sterile filtration, aseptic filling, terminal sterilisation if applicable, drug-product lyophilisation, container-closure integrity, visual inspection and batch release require a separately named and qualified third party.

3.3 Device, secondary packaging and distribution

No current Ajinomoto Japan combination-device assembly, pen/autoinjector manufacture, primary-container manufacture or commercial secondary-packaging capability was evidenced for peptide products. PCI's device/packaging services cannot be attributed to Ajinomoto after divestment.[12] Labelling, serialization, tamper evidence, South African pack configuration and legal release should be placed in a separate workstream.

3.4 Research-only and non-GMP services

Osaka offers research-to-commercial services and explicitly lists research-support activities.[2] Its solid-phase page distinguishes non-GMP from cGMP output and lists applications including basic drug discovery, animal experiments, diagnostics and development-stage material.[7] Such lots are ****not eligible for regulated therapeutic supply**** without documented GMP manufacture and full comparability/transfer. CORYNEX strain/process/assay work in Japan is also development work until transferred to a named GMP CMO.[15]

4. Regulatory evidence and market mapping

| Authority/market | Evidence found by decision date | Assessment |

|---|---|---|

| ****United States — FDA product approval**** | Ajinomoto announced a first US-approved commercial drug made by AJIPHASE in October 2020.[9] The product and application are undisclosed; context identifies it as an oligonucleotide, not a proven peptide. | Evidence that the platform has supported one regulated commercial dossier, ****not evidence of an FDA-approved Ajinomoto-manufactured peptide****. Do not infer the customer. |

| ****United States — FDA inspections**** | FDA's primary February 2025 Form 483 names Ajinomoto Co., Kawasaki Plant, FEI 3002806336, as an API manufacturer.[10] A secondary FDA-data compilation reports 2025 VAI, 2019/2017/2015 NAI and earlier VAI outcomes.[11] | Corporate/API quality evidence, but wrong site for Tokai/Osaka peptide qualification. Obtain the final classification letter/EIR and site-specific Tokai/Osaka FDA inspection history directly. FDA warns its public dashboard is not comprehensive.[17] |

| ****FDA warning letters/import alerts**** | No Ajinomoto Tokai, GeneDesign/Osaka or Kawasaki human-drug warning letter or GMP import-alert listing was located in the reviewed public searches. | ****Not proof of absence.**** Require signed disclosure of all inspections, 483s, EIRs, warning/untitled letters, recalls, import actions and commitments for at least ten years. |

| ****Japan — PMDA/MHLW**** | Japanese facilities and cGMP claims are public, but no publicly retrievable site-specific PMDA/MHLW GMP result, manufacturing-business licence, product review or approved peptide naming Tokai/GeneDesign was located. | Domestic location does not establish a particular product approval or inspection outcome. Obtain MHLW/competent-prefecture licences and product-specific GMP compliance documentation. |

| ****EU/EEA — EudraGMDP/EMA**** | No public Tokai/GeneDesign peptide GMP certificate, non-compliance statement

or identified EMA dossier naming these sites was located. | Market/service territory for peptide API ****not publicly disclosed****. A current EudraGMDP extract and EU QP declaration, if applicable, are

RFI items. |

| ****EDQM CEP**** | No peptide CEP tied to Tokai or GeneDesign was located. | CEP status ****not publicly evidenced****; absence is not necessarily adverse because custom peptide APIs may be supported by ASMF/DMF rather than a CEP. |

| ****TGA, Health Canada, WHO, NMPA, MFDS**** | No site-specific therapeutic-peptide regulator record was located in the reviewed public searches. | Countries/markets served for peptides ****not publicly disclosed****. Do not infer from the group's global offices or customer statements. |

| ****South Africa — SAHPRA**** | No SAHPRA registration, GMP recognition or South African product label naming Ajinomoto Japan peptide manufacture was located. | Never describe the company or site as SAHPRA-approved. A product-specific SAHPRA pathway, foreign-site GMP evidence and local applicant responsibilities remain to be established. |

Only the ****United States**** can be mapped from an identified approval claim, and that evidence is for an undisclosed AJIPHASE drug apparently in the oligonucleotide modality. It does not establish a commercial peptide market. Japanese operations and Japanese customers mentioned historically are not equivalent to a peptide product approval. No customers should be inferred or named.

5. Quality, testing and data-integrity assessment

Ajinomoto's public quality pages claim an electronic quality system, defined SOPs, internal audit, GMP-qualified vendors, calibration, validation, QC/QA/document-control personnel, global compliance, operational-metric trending, management review, regulatory response and maintenance of CDER/CBER master files.[18][19] These statements are useful RFI starting points but are ****corporate marketing assertions****, not site-specific evidence of execution.

The peptide/oligonucleotide analytical page lists RP-HPLC, IEX-HPLC, MALDI-TOF/MS, residual-salt testing/removal and endotoxin testing, and says raw materials, bulk solutions, in-process materials and finished product are characterised.[20] This supports basic identity/purity and process-control capability. It does not publicly establish peptide-specific validated release and stability panels, method validation/transfer packages, potency/biological activity where relevant, water and counter-ion determination, residual solvents, elemental impurities, amino-acid analysis, chiral purity/racemisation, sequence confirmation by LC-MS/MS, deletion/truncation impurity mapping, aggregation, bioburden, microbial limits, particulate control, reference-standard qualification or stability-indicating forced-degradation studies.

Osaka's published solid-phase table includes HPLC or MPLC purification, ultrafiltration desalting and vial/bulk lyophilisation up to a stated 10 kg—but the page is explicitly about ****oligonucleotides****. [7] It must not be used as proof of peptide purification or peptide bulk lyophilisation. Tokai peptide downstream equipment and final-form capability remain to be documented.

The public quality pages do not describe site-specific OOS/OOT procedures, computerized-system validation, audit trails, ALCOA+ controls, chromatography-data governance, deviation/CAPA effectiveness checks, change control, product-quality reviews, reserve samples or stability chambers. FDA's 2025 Kawasaki observations make these RFI questions more important: five reviewed deviation/OOS investigations did not identify the true root cause or adequate actions to prevent recurrence; laboratory worksheets lacked issuance control; and material-compatibility evidence for API tanks was deficient.[10] FDA describes a 483 as inspectional observations rather than a final compliance determination, and final NAI/VAI/OAI classification is a separate agency step.[17] The observations cannot be generalized to Tokai or GeneDesign, but corporate governance should demonstrate how lessons learned and CAPA were propagated across Japanese pharmaceutical sites.

The former San Diego affiliate's January 2023 Form 483 cited inadequate quality oversight of validation, unqualified QC-material storage, deficient disinfectant efficacy/cleaning records and facility maintenance, and delayed cGMP logbook review.[16] Because Althea was sold, this is

****historical brand/ownership context****, not current Tokai/Osaka status. It nevertheless warrants asking how global quality oversight worked before and after the divestment.

6. Adverse history and unresolved searches

No public evidence reviewed showed an FDA OAI classification, warning letter, drug-GMP import alert, EMA non-compliance statement, EDQM suspension, WHO notice or Japanese administrative sanction specifically against Tokai AJIPHASE or GeneDesign Osaka. The conclusion is strictly ****“not located in the public sources reviewed,” not “none exists.”**** FDA itself states that its inspection database is incomplete and excludes some inspection categories and pending actions.[17]

The known 2025 Kawasaki VAI/483 is material because it concerns an Ajinomoto Co. API operation and touches OOS, deviations, CAPA, document control and equipment-material suitability.[10][11] The exact APIs and whether any peptide-related operation was in scope are redacted/not established. The 2023 Althea 483 is material only as former-affiliate history.[16]

No public EIR was located for Tokai or Osaka, and no EIR was retrieved for Kawasaki. The buyer should request all EIRs, final classification letters and written responses/CAPA evidence, plus a complete regulator inspection table by legal entity, site, product and system inspected.

7. Supply-chain and operational dependencies

| Dependency | Evidence-based concern | Required control |

|---|---|---|

| ****Protected amino acids, coupling/deprotection reagents and AJIPHASE anchor**** | AJIPHASE uses an anchor, organic/polar solvents, excess reagents and filtration.[5][13] Supplier names, countries, dual-source status, specifications and change-notification terms are not public. | Bill of materials; TSE/BSE and animal-origin statements; nitrosamine and mutagenic-impurity assessments; chirality controls; supplier audits; dual-source/stock policy; anchor IP and sole-source analysis. |

| ****Resin and solid-phase inputs**** | AJIPHASE reduces reliance on resin, but early-stage/Osaka work uses solid-phase methods.[6][7][13] | Identify resin/synthesizer consumables by stage and whether process transfer from SPPS to AJIPHASE changes impurity profiles. |

| ****Solvents and waste handling**** | The platform claims lower solvent/reagent use, not elimination.[5][9] Large-volume operations remain dependent on solvent availability, recovery, hazardous logistics and disposal. | Solvent sourcing, recycled-solvent policy, residual-solvent specifications, recovery validation, outage inventory and environmental permit confirmation. |

| ****Chromatography, membranes and analytical columns**** | Public methods include HPLC/MPLC, UF, RP/IEX-HPLC and MALDI-TOF.[7][20] Peptide-specific use and supplier resilience are undisclosed. | Resin/column/membrane lifecycle, carryover controls, cleaning validation, spare strategy and extractables/leachables. |

| ****Single-site concentration**** | Tokai is the only publicly identified Japanese large-scale AJIPHASE peptide site.[4] | Business-continuity plan, alternate train/site, disaster recovery, utilities redundancy, tech-transfer package, insurance and inventory buffers. Japan-specific seismic/typhoon risk should be evaluated without treating an event as inevitable. |

| ****CORYNEX subcontracting**** | Japanese team performs R&D; partner CMOs perform GMP production. The Olon partnership adds large-scale fermentation.[14][15] | Name every CMO/site; quality and technical agreements; audit rights; deviation/change notification; data ownership; cell-bank control; release responsibilities; regulatory filin

g consistency; no unapproved subcontracting. |

| ****Sterile fill and finished dose**** | Current Ajinomoto sterile affiliate was divested.[12] |

Independently source and qualify drug-product manufacturer; validate shipping from Japan/partner DS site; establish QP/import testing where relevant and SAHPRA dossier roles. |

| ****Cold chain and transport**** | No public peptide bulk storage temperature, hold-time, freeze/thaw or shipping-validation evidence was found. | Molecule-specific thermal/light/oxygen sensitivity, bulk

container compatibility, excursion management, lane qualification to South Africa, backup freight and customs plans. |
| ****Device/packaging**** | No current in-scope capability evidenced. | Separate device, primary-container and packaging suppliers; combination-product design control; container-closure integrity; serialization and South African labelling. |
| ****Geopolitical/logistics**** | Japan-to-South Africa supply entails long lanes and may depend on imported monomers, solvents, columns or equipment parts; source countries are undisclosed. | Country-of-origin map, sanctions/export-control review, port/airfreight alternatives, currency and lead-time buffers. Do not assume China or any other country without supplier disclosure. |

Ajinomoto claims a “broad base of GMP qualified vendors,” which may reduce procurement risk, but no supplier list, qualification evidence or commodity-level redundancy is public.[18] The claim must be tested against the project-specific bill of materials.

8. South African fit and gating RFI

Ajinomoto Japan is a ****credible candidate for peptide drug-substance RFI****, particularly where a synthetic peptide is suitable for AJIPHASE or a long peptide/protein is suitable for CORYNEX. It is not publicly evidenced as an integrated South African-ready drug-product supplier. SAHPRA-facing suitability will depend on product-specific GMP evidence, a complete API CMC package/DMF or equivalent letter of access, inspection and CAPA transparency, stability data supporting the intended South African climatic/lane conditions, and separately qualified fill-finish, packaging and device partners.

Before any approval, the buyer should obtain and verify:

1. Current Japanese registry extracts, manufacturing-business/manufacturing-site licences and an organisation chart linking Ajinomoto Co., Tokai, Tokyo and GeneDesign.
2. Exact proposed legal manufacturer and site/building/train; subcontractors; testing laboratories; release site; warehousing and shipping points.
3. Product-specific regulatory history, DMF/ASMF/CEP status, agency inspection table, EIRs/classification letters, 483 responses and CAPA effectiveness evidence.
4. At least three representative GMP peptide batches at proposed scale, process performance/yield, impurity fate/purge, specifications, certificates of analysis and deviation/OOS history.
5. Peptide-specific analytical validation, reference standards, stability protocols/data, forced degradation, mass balance, sequence and stereochemical confirmation, residual solvent/reagent/elemental impurity controls, bioburden/endotoxin where applicable and method-transfer plans.
6. Raw-material supplier/country and dual-source map for protected amino acids, anchor/resin, coupling reagents, solvents, chromatography media and membranes; change control and shortages history.
7. Data-integrity architecture, audit-trail review, LIMS/CDS/ERP validation, paper worksheet control, OOS/OOT/deviation/CAPA procedures and evidence that Kawasaki 2025 observations were assessed across relevant sites.
8. Cleaning validation, cross-contamination and carryover controls, equipment material compatibility/extractables-leachables, hold times, bulk container and shipping validation.
9. A named sterile fill-finish/drug-product, packaging/device and South African import/release chain with independent qualification.
10. Business-continuity, disaster recovery, cybersecurity, cold-chain and lane-excursion plans, plus termination

/technology-transfer rights.

9. Preliminary tier rationale

****Tier B — credible but material gaps.**** The company merits an RFI because the legal identity and

parentage are substantially resolvable; Tokai has a specific large-scale AJIPHASE proposition; Osaka has established cGMP oligonucleotide infrastructure and peptide-development listings; AJIPHASE has supported an undisclosed FDA-approved commercial oligonucleotide; and public FDA inspection records exist for a separate Ajinomoto API site. Tier A is not justified because public evidence does not connect a named therapeutic peptide, its dossier and regulated-market approval to Tokai or Osaka; no site-specific Tokai/Osaka regulator inspection classification or GMP certificate was verified; critical peptide-scale, purification, lyophilisation, testing, stability, data-integrity and CAPA details remain marketing-level; CORYNEX commercial GMP relies on partners; and sterile fill-finish/device/packaging are outside the current in-group offering.

This assessment does ****not**** state that Ajinomoto, any site or any product is fully compliant, SAHPRA-approved or safe.

References

[1]: Ajinomoto Bio-Pharma Services, "Ajinomoto Althea and OmniChem Combine to Form Ajinomoto Bio-Pharma Services," 2 October 2018. <https://www.ajibio-pharma.com/news-post/ajinomoto-althea-and-omnichem-combine-to-form-ajinomoto-bio-pharma-services-reflecting-a-collective-range-of-services-offered-to-the-bio-pharma-industry/>

[2]: GeneDesign, Inc./Ajinomoto Bio-Pharma Services, "Specialists in Oligonucleotides" (current company information accessed for decision date). <https://www.ajioligos.com/en/about/specialists-in-oligonucleotides/>

[3]: Ajinomoto Co., Inc., "Global Network — Ajinomoto Bio-Pharma Services Osaka (GeneDesign, Inc.)." https://www.ajinomoto.com/aboutus/global_network

[4]: Ajinomoto Bio-Pharma Services, "Japan" (Tokyo, Tokai and Osaka site roles and addresses). <https://www.ajibio-pharma.com/contact-us/japan/>

[5]: Ajinomoto Bio-Pharma Services, "AJIPHASE Synthesis." <https://www.ajibio-pharma.com/cdmo-services/oligo-peptide-synthesis/ajiphase-synthesis/>

[6]: Ajinomoto Bio-Pharma Services, "Oligo & Peptide Synthesis." <https://www.ajibio-pharma.com/cdmo-services/oligo-peptide-synthesis/>

[7]: Ajinomoto Bio-Pharma Services, "Solid Phase Synthesis." <https://www.ajibio-pharma.com/cdmo-services/oligo-peptide-synthesis/solid-phase-synthesis/>

[8]: Ajinomoto Bio-Pharma Services, "GeneDesign Joins Ajinomoto Bio-Pharma Services and Opens Oligonucleotide API Development Center," 9 April 2019. <https://www.ajibio-pharma.com/news-post/genedesign-joins-ajinomoto-bio-pharma-services-and-opens-oligonucleotide-api-development-center/>

[9]: Ajinomoto Bio-Pharma Services, "First Commercial Drug Manufactured via Ajinomoto Bio-Pharma Services' AJIPHASE Technology Receives FDA Approval," 6 October 2020. <https://www.ajibio-pharma.com/news-post/first-commercial-drug-manufactured-via-ajinomoto-bio-pharma-services-ajiphase-technology-receives-fda-approval/>

[10]: U.S. Food and Drug Administration, Form FDA 483, Ajinomoto Company Inc., Kawasaki Plant, FEI 3002806336, inspection 17–21 February 2025. <https://www.fda.gov/drugs/cder-foia-electronic-reading-room/ajinomoto-company-inc-02212025>

[11]: ProPublica, Rx Inspector, "Ajinomoto Company, Inc." (compilation of FDA inspection data; secondary source). <https://projects.propublica.org/rx-inspector/facilities/3002806336/>

[12]: Ajinomoto Co., Inc./Ajinomoto Bio-Pharma Services, "Ajinomoto Co., Inc. to Sell its Entire

Equity Stake in Ajinomoto Althea, a US-based Aseptic Fill/Finish CDMO" (completed 1 May 2025). <https://www.ajibio-pharma.com/news-post/ajinomoto-co-inc-to-sell-its-entire-equity-stake-in-ajinomoto-althea-a-us-based-aseptic-fill-finish-cdmo/>

[13]: Ajinomoto Co., Inc., "Ajinomoto Group Acquires All Shares of GeneDesign, Inc.," 9 December 2016. https://www.ajinomoto.co.jp/company/en/presscenter/press/detail/g2016_12_09.html

[14]: Ajinomoto Bio-Pharma

Services, "Ajinomoto Bio-Pharma Services and Olon S.p.A. Form Strategic Partnership to Advance Large-Scale Peptide and Protein Manufacturing," 28 October 2025. <https://www.ajibio-pharma.com/news-post/ajinomoto-bio-pharma-services-and-olon-s-p-a-form-strategic-partnership-to-advance-large-scale-peptide-and-protein-manufacturing/>

[15]: Ajinomoto Co., Inc., "CORYNEX — Secretion production of peptides/proteins by gram-positive bacterium." <https://ajibio-pharma.ajinomoto.com/corynex/>

[16]: U.S. Food and Drug Administration, Form FDA 483, Ajinomoto Althea, Inc. d/b/a Ajinomoto Bio-Pharma Services, FEI 3004575449, issued 16 January 2023. <https://www.fda.gov/about-fda/fda-commissioner/ajinomoto-althea-inc-dba-ajinomoto-bio-pharma-services-san-diego-ca-483-issued-01162023>

[17]: U.S. Food and Drug Administration, "Inspection Classification Database" (NAI/VAI/OAI definitions and coverage limitations), content current 13 September 2024. <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/inspection-classification-database>

[18]: Ajinomoto Bio-Pharma Services, "Quality Management." <https://www.ajibio-pharma.com/dedication-to-quality/quality-management/>

[19]: Ajinomoto Bio-Pharma Services, "Regulatory Excellence." <https://www.ajibio-pharma.com/dedication-to-quality/regulatory-excellence/>

[20]: Ajinomoto Bio-Pharma Services, "Product Quality & Analytical Testing." <https://www.ajibio-pharma.com/cdm-services/oligo-peptide-synthesis/product-quality-analytical-testing/>

[21]: Ajinomoto Bio-Pharma Services, "Process Development." <https://www.ajibio-pharma.com/cdm-services/oligo-peptide-synthesis/process-development/>

09-M09.md

HLB PEP Co., Ltd. — Therapeutic-Peptide Manufacturer/CDMO Due-Diligence Dossier

****Dossier ID:**** 09-M09

****Decision date / evidence cut-off:**** 20 August 2026

****Jurisdiction:**** Republic of Korea (South Korea)

****Purpose:**** Preliminary desk prequalification for regulated therapeutic supply to a South African business

****Preliminary desk tier:**** ****B** — credible manufacturer, but material regulatory and quality-system gaps require RFI and audit closure before qualification

1. Executive conclusion

HLB PEP Co., Ltd. is a KOSDAQ-listed South Korean peptide company, formerly ****AnyGen Co., Ltd.****, with two disclosed MFDS-GMP peptide manufacturing plants and a separate non-GMP/research operation. Its strongest externally verifiable regulated-market evidence is: (i) Korean API product authorisations and MFDS GMP status disclosed in current securities filings; (ii) three ****active US FDA Type II DMFs****—leuporelin acetate DMF 25878, ganirelix acetate DMF 40573, and arginine-vasopressin acetate DMF 44390—in FDA's official list current to 30 June 2026; and (iii) Japanese foreign-manufacturer recognition reported as renewed in August 2025.^{[1][2]} The company is therefore credible enough for a formal request for information (****RFI****), but the public record is not strong enough for Tier A.

The central caution is a ****one-month MFDS manufacturing suspension in 2025**** covering desmopressin acetate and leuporelin acetate made at the Jangseong plant because annual product-quality-review reports were not prepared for those products in 2022 and 2023. The company states that it began preparing the reports for all manufactured products in 2024, but no regulator closure letter, effectiveness check, root-cause record, retrospective product-impact assessment, or CAPA package was found publicly.^[3] This directly raises questions about pharmaceutical quality-system oversight, trend review, deviation/OOS aggregation, management review, and data governance.

No public evidence located by the cut-off establishes an FDA inspection, FDA inspection classification (NAI/VAI/OAI), EIR, FDA 483, US approved-product dependency on an HLB PEP site, EMA/EudraGMDP certificate, EDQM CEP, WHO prequalification, TGA GMP clearance, Health Canada establishment evidence, or SAHPRA registration/approval. FDA's inspection database is explicitly non-comprehensive and omits pre-approval inspections, so absence from the public search is not proof that no FDA activity occurred.^[4] The February 2025 Osong event promoted by the company was a ****private mock inspection by Interchem****, not an FDA inspection or cGMP “approval”; the report itself says CAPA observations remained to be addressed before a future FDA inspection.^[5]

The evidenced offering is ****peptide API/drug substance and associated development/analytical services****. Public sources do ****not**** substantiate sterile finished-dose manufacture, aseptic fill-finish, combination-device assembly, primary-container closure operations, or commercial finished-dose packaging. Website “lyophilization” is API isolation/drying and must not be interpreted as sterile vial lyophilization.^[6]

> ****Decision:**** Maintain at ****Tier B**** and issue a site- and product-specific RFI. Do not approve for South African therapeutic supply until the proposed API/site is identified; current MFDS GMP evidence, full inspection history, the 2025 sanction CAPA and effectiveness evidence, data-integrity controls, stability/retest program, impurity strategy, supplier qualification, transport validation, and any subcontracting are reviewed. Sterile fill-finish and device/packaging must be sourced and qualified separately unless HLB PEP supplies auditable evidence beyond the public record.

2. Legal entity, ownership and corporate continuity

The current legal and commercial name is **■■■■■■■ ■■■■■ / HLB Pep Co., Ltd.** The 2026 first-quarter filing gives establishment on 2 May 2000, registered head office at Room 206, Pilot Production Building,

Gwangju Technopark, 333 Cheomdangwagi-ro, Buk-gu, Gwangju, and KOSDAQ code 196300. The filing states that the name changed from ****AnyGen Co., Ltd.**** to ****HLB PEP Co., Ltd.**** in March 2025; the company website lists business registration number ****409-81-53794****.^[1]^[7]

HLB Genex, Inc. became the largest shareholder through a third-party allotment in March 2025. A 10 July 2026 statutory beneficial-ownership filing identifies **HLB Genex, Inc.** as HLB PEP's largest shareholder and reports 5,000,262 shares and share-equivalent instruments, or 42.70%, held by HLB Genex and concert parties; actual voting common shares represented 28.44% in that filing. HLB Co., Ltd. is the largest shareholder of HLB Genex. This supports describing HLB PEP as an HLB Group controlled/affiliated company, but **HLB PEP remains the manufacturing legal entity** and has no disclosed consolidated subsidiaries in its 2026 first-quarter report.[^8][^1]

| Item | Resolved position at 20 August 2026 |

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| Legal manufacturer / contracting entity | ****HLB Pep Co., Ltd.****, formerly AnyGen Co., Ltd. |

| Korean name | ■■■■■■ ■■■■ |

| Business registration number | 409-81-53794 |

| Listed identity | KOSDAQ 196300; FDA SRO issuer record also identifies HLB Pep Co., Ltd., South Korea, ticker 196300 and DUNS 689867138. The SRO record is an issuer identity record, ****not**** an establishment registration, inspection or product approval.[^9] |

| Largest shareholder / group | HLB Genex, Inc.; HLB Co., Ltd. is HLB Genex's largest shareholder.[^8] |

| Subsidiaries | None disclosed as consolidated subsidiaries in the 2026 Q1 filing.[^1] |

3. Sites and roles

The official location page identifies five Korean locations.^[10] Only the two factories are treated as potentially relevant to regulated therapeutic manufacture; the Gwangju operation must be segregated as non-GMP/research unless batch-specific evidence proves otherwise.

| Site | Publicly stated role | Therapeutic-supply interpretation |

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| **Jangseong Plant 1**, 123 Nanosandan-ro, Nam-myeon, Jangseong-gun, Jeollanam-do 57248, inside Nano Bio Research Center | GMP peptide manufacturing plant. The company reported MFDS GMP re-certification in July 2025. Historical company material describes one line and an anti-cancer peptide API focus. Desmopressin acetate and leuporelin acetate manufactured here were subject to the 2025 suspension.[^1][^3][^11] | Potential peptide API/drug-substance site only. Product/site scope must be confirmed from the current MFDS certificate and manufacturing authorisation. The 2025 APQR failure is directly site relevant. |

| **Osong Plant 2** | 773 Jeongjungyeonje-ro, Osong-eup, Heungdeok-gu, Cheongju-si, Chungcheongbuk-do 28161 | Purpose-built peptide plant completed in 2018; company reports MFDS GMP re-certification in January 2026. Historical company material describes three manufacturing lines and non-oncology peptide drug-substance work. The February 2025 event was a private mock FDA inspection of leuporelin and ganirelix systems.[^1][^5][^11] | Potential peptide API/drug-substance and clinical-material site. No public FDA inspection/classification was located. Site transfer/change-control and cross-site comparability require confirmation. |

| **Gwangju headquarters / pilot operation**, Room 206, Pilot Production Building, Gwangju Technopark, 333 Cheomdangwagi-ro, Buk-gu, Gwangju 61008 | Registered head office. The 2023 audited filing separately reports “Gwangju Non-GMP” capacity measured in jobs, not grams.[^10][^12] | **Research/custom/catalogue and non-GMP work must be excluded from

regulated therapeutic supply.** Clarify physical and electronic segregation from GMP operations. |
| **Corporate research institute**, Room 403, Industry-Academic Cooperation Research Building,
GIST, 123 Cheomdangwagi-ro, Buk-gu, Gwangju 61005 | Research institute.[^10] | R&D only on
current public evidence; not a qualified therapeutic manufacturing site. |
| **Seoul sal

es office**, Room 821, Creation Business Center, 8F Samjin Building, 113 Ahasan-ro,
Seongdong-gu, Seoul 04794 | Sales office.[^10] | No manufacturing role. |

The public record does not resolve whether analytical release laboratories are duplicated at both
GMP plants, whether any testing is conducted at Gwangju/GIST, which site owns each DMF, or
whether warehousing and final API packaging occur at Jangseong, Osong, or a third party. These
must be answered in the RFI.

4. Service-scope classification

| Service category | Finding | Evidence status |

| --- | --- | --- |

| **Peptide API / drug substance** | **Yes.** Company and filing evidence supports chemical
peptide synthesis, preparative HPLC purification, API isolation/lyophilisation, QC and regulatory
support. Korean API registrations and FDA Type II DMFs substantiate regulated drug-substance
activity.[^1][^2][^6] | Verified at entity/portfolio level; product-to-site mapping remains incomplete. |

| **Clinical peptide drug substance / CDMO** | **Yes, with qualification caveat.** Current filings
describe investigational-drug peptide production, process development, specifications and
analytical-method development; earlier audited disclosure lists CMC, stability testing and
clinical/nonclinical custom manufacture.[^12][^13] | Credible filing evidence, but no customer
identity is inferred and no sponsor approval is treated as site approval. |

| **Sterile finished dose** | **Not publicly evidenced.** | No finished-dose dosage-form line, sterile
product licence, approved label, or regulator record located. |

| **Aseptic fill-finish / sterile vial lyophilisation** | **Not publicly evidenced.** Website lyophilisation
concerns isolation of bulk peptide API in trays under nitrogen; it is not evidence of aseptic filling or
sterile container closure.[^6] | Treat as out of scope. |

| **Device / combination product / packaging** | **Not publicly evidenced.** | No syringe, pen,
pump, device assembly, primary packaging, serialization, or finished-pack line found. |

| **Research-only catalogue/custom peptides** | **Yes, separate and material.** The website
explicitly calls catalogue peptides research-use products, and the filing reports 41.31% of Q1 2026
revenue from research-use peptide materials.[^1][^7] | Must be excluded from therapeutic
consideration and segregated from GMP sourcing. |

| **Cosmetic peptides** | Yes, marketed separately. | Not relevant to therapeutic qualification. |

5. Peptide manufacturing capabilities

5.1 Synthesis and scale

HLB PEP's website claims more than 20 years' peptide-synthesis experience and describes **Fmoc
solid-phase peptide synthesis (SPPS)** for multi-kilogram quantities, solution-phase peptide
synthesis for multi-gram to multi-kilogram lots, and convergent fragment synthesis combining SPPS
and solution coupling.[^6] Its March 2026 shareholder material describes a strategy to increase
plant capacity to **500–1,000 g per batch** and states, in general industry discussion, that
competitive peptide API production requires kilogram-scale synthesis and purity above 97%.[^13]
These are company representations, not independently witnessed batch records.

Audited 2023 disclosure provides a more conservative operational metric: Jangseong annual
capacity 4,935 g with 4,494 g produced (91.06% utilisation), and Osong annual capacity 2,620 g
with 941 g produced (35.91% utilisation). The disclosed capacity formula was
labour-based—headcount × workdays × output per person—not reactor/synthesiser nameplate
capacity. It therefore does not independently demonstrate multi-kilogram batch capability.[^12] The

RFI should reconcile website “multi-kg” claims, the 500–1,000 g/batch expansion target, actual validated maximum scale by peptide, and 2024–2026 capacity/utilisation.

5.2 Purification and isolation

The official capability page describes preparative HPLC columns of 8–20 cm internal diameter and 50 cm length, ODS C18 10 µm packing, and claimed purifi

cation throughput of 350–500 g/day. It describes a classified isolation suite, stainless-steel 316L 15-tray lyophiliser connected to nitrogen purge, 50 kg ice capacity, and gram-to-multi-kilogram API quantities per batch. The page says systems are validated.^[6] No public validation summaries, cleaning limits, carryover studies, resin/column lifecycle controls, solvent-recovery controls, or viral/TSE assessments were located.

5.3 Facilities and environmental controls

The same page claims Class 100,000 to Class 10,000 cleanrooms and, for purification/isolation, a Class 100 booth within a Class 100,000 room.^[6] The terminology appears to use legacy Federal Standard classes; the corresponding ISO classification, viable/non-viable monitoring program, alarm/action limits, HVAC qualification, differential-pressure cascade and campaign changeover controls should be requested. These classifications do not by themselves establish aseptic processing or sterility.

6. API portfolio and regulatory evidence

6.1 FDA DMFs

FDA's official Q2 2026 DMF list is current through DMF 044443 and includes submissions received by 30 June 2026 for which acknowledgement letters were sent before 19 July 2026.^[2] Direct review of the downloadable FDA spreadsheet identified:

DMF	Holder in FDA list	Subject	Status at 30 June 2026	Interpretation
25878	ANYGEN CO LTD	Leuprorelin acetate	**A — active**	Strong dossier-presence evidence; filed 12 Dec 2011. An active DMF is not an FDA approval and does not prove current site inspection status.
40573	ANYGEN CO LTD	Ganirelix acetate	**A — active**	Strong dossier-presence evidence; filed 17 Oct 2024.
44390	HLB PEP CO LTD	Arginine-vasopressin acetate	**A — active**	Filed 29 Jun 2026. The holder-name transition supports legal continuity but must be confirmed in letters of access and DMF administrative records.

No other HLB PEP/AnyGen entry was identified in that official list by exact holder-name search. FDA describes DMF status as active/inactive; a DMF is confidential support information and not a stand-alone product approval. The company website also presents desmopressin as “DMF,” but the exact jurisdiction/number was not substantiated as a US DMF in FDA's current list.^{[2][14]}

6.2 Korea (MFDS)

Current securities filings report MFDS product authorisations for leuprorelin (2011), desmopressin (2012), ziconotide (2013), ganirelix (2023), and vasopressin (November 2025), plus export-only authorisations for buserelin and goserelin in 2017. They also report Jangseong MFDS GMP re-certification in July 2025 and Osong re-certification in January 2026.^{[1][15]} Publicly accessible certificate numbers, scopes, expiry dates and site/product annexes were not located and must be obtained directly.

An MFDS English product page dated 20 May 2016 lists AnyGen's leuprolide, desmopressin,

exenatide, octreotide and ziconotide acetates and states exporting countries as “Japan, China, South Asia etc.”^[16] This is regulator-hosted historical export evidence but is not a current country-by-country approval list and should not be extrapolated to named customers.

6.3 Japan, India and other jurisdictions

The 2026 filing reports Japanese MHLW foreign-manufacturer recognition renewed in August 2025 and a 2016 Indian DCGI export authorisation for leuprorelin.^[1] Certificate/approval numbers and current scopes were not publicly retrieved. These are credible issuer filings but require primary certificate copies and confirmation that they cover the proposed product and site.

The company's history states that a leuprorelin DMF application was made to China's then-CFDA in 2014, but an application is not evidence of registration or market supply.^[15] No current NMPA registration, EMA/EudraGMDP record, EDQM CEP, TGA clearance, Health Canada establishment record, or WHO prequalification was located.

6.4 Evidenced market map

| Market

/ region | Evidence | What may be concluded |

| --- | --- | --- |

| **South Korea** | MFDS API authorisations and current GMP re-certifications reported in current securities filing.^[1] | Domestic regulated API activity is evidenced, subject to certificate verification and the 2025 sanction. |

| **United States** | Three active Type II DMFs in FDA's official Q2 2026 list.^[2] | US dossier presence only; no FDA-approved finished product, inspection classification, EIR or commercial supply relationship is established. |

| **Japan** | MHLW foreign-manufacturer recognition reportedly renewed Aug 2025; MFDS 2016 page lists Japan among export destinations.^{[1][16]} | Regulatory/export linkage is evidenced, but current product/site scope and actual ongoing supply are not public. |

| **India** | DCGI leuprorelin export authorisation reported in filing.^[1] | Product-specific regulatory linkage; current validity and sales are not public. |

| **China / South Asia (region)** | MFDS 2016 product page states export destinations “Japan, China, South Asia etc.”^[16] | Historical export indication only; no current NMPA approval or named South Asian country should be inferred. |

| **EU/EEA, Australia, Canada, WHO/PQ, South Africa** | No qualifying public regulator/product evidence located. | **Not publicly disclosed/evidenced.** |

No customers are inferred. Initials or unnamed counterparties in filings are not decoded.

7. Testing, release, stability and impurity control

The website lists standard API specifications including appearance, solubility, identity by mass spectrometry/amino-acid analysis/HPLC, counterion, moisture, peptide content by nitrogen, trifluoroacetic acid, chloride, residual organic solvents, bioburden, endotoxin and product-specific parameters. Listed equipment includes MALDI-TOF MS, FT-IR, UV/Vis, polarimeter, Karl Fischer, HPLC, GC, TOC, ELISA reader and biological safety cabinet.^[6] Earlier audited disclosure lists CMC work, stability testing, process development, QC method development and QA-process development as service offerings.^[12]

These disclosures are useful but incomplete for regulated therapeutic qualification. No public method-validation packages, specifications/CoAs, ICH stability protocols, retest periods, forced-degradation studies, elemental-impurity assessments, nitrosamine assessments, genotoxic-impurity controls, amino-acid racemisation controls, deletion/truncation sequence mapping, aggregation controls, oxidation/deamidation controls, residual-coupling-reagent limits, chromatographic integration governance, reference-standard qualification, microbial method

suitability, or batch-trend data were found.

For peptide-specific impurity control, request at minimum: full related-substance map and fate/purge studies; LC-MS/MS or orthogonal structural characterisation; control of deletion/insertion sequences, epimers, oxidised/deamidated species and counterion forms; residual TFA and solvent strategy; residual protected amino acids/coupling reagents/scavengers; resin-derived leachables; peptide-content/potency basis; and justification against USP/Ph. Eur. monographs where claimed. The website's pharmacopoeial and DMF labels are marketing summaries and do not replace current specifications or regulator correspondence.^[14]

8. Pharmaceutical quality system, data integrity, OOS/OOT and CAPA

Public sources provide no substantive procedure-level evidence for data integrity, ALCOA+ controls, audit trails, user-access management, backup/disaster recovery, electronic raw-data retention, chromatographic integration review, deviation handling, OOS/OOT investigations, change control, complaint/recall systems, or supplier CAPA.

The strongest negative signal is the 2025 MFDS administrative action. HLB PEP states that the Gwangju Regional MFDS ordered a one-month manufacturing suspension for Jangseong-produced desmopressin acetate and leuporelin acetate because annual quality-evaluation reports were not prepared in 2022 and 2023. It says all products have received such reports since 2024 and that inventory would avoid customer interruption.^[3] This is an acknowledged GMP-system lapse across two years, not merely a documentary typo. Public evidence does not resolve whether the omitted reviews concealed recurring deviations, OOS/OOT trends, complaints, stability drift, supplier changes or process-capability deterioration.

The February 2025 private mock inspection at Osong reviewed manufacturing-process effectiveness, environment/facilities, quality control, calibration and maintenance. The company's own report says the consultant identified corrective and preventive action points to be remediated before a future FDA inspection.^[5] No observation list or CAPA closure evidence is public. This event must not be represented as an FDA inspection, FDA pre-approval inspection, EIR, classification, or approval.

Mandatory quality RFI evidence

1. The MFDS administrative disposition, inspection report, exact suspension dates, impacted licence/product numbers, root-cause analysis, retrospective product-impact/risk assessment, CAPA plan, effectiveness checks and regulator closure.
2. APR/PQRs for every proposed product for 2022–2025 and cross-product trend reports, including deviations, OOS/OOT, complaints, recalls, yields, impurity trends, stability, changes, rejected/reprocessed batches and CAPA effectiveness.
3. Full regulatory inspection history for both sites since 2019, including MFDS, PMDA/MHLW, FDA or other authorities; 483s, EIRs/classification letters, observations and responses.
4. Data-integrity risk assessment; validated-system inventory; audit-trail review; CDS/LIMS/ERP controls; privileged-user governance; backup/restore validation; raw-data retention; and any data-integrity investigations.
5. OOS/OOT/deviation/change-control/CAPA SOPs and three recent anonymised examples demonstrating scientifically sound root cause and effectiveness review.
6. Quality agreement template, release authority, deviation notification timelines, right to audit, regulator-contact notification, change notification, and subcontractor governance.

9. Adverse regulatory history and negative searches

| Topic | Finding by 20 Aug 2026 |

| --- | --- |

| MFDS administrative action | ****Confirmed:**** one-month Jangseong manufacturing suspension for

desmopressin acetate and leuporelin acetate due to missing 2022–2023 annual quality-evaluation reports.^[3] |

| FDA inspections/classification/EIR/483 | No qualifying public record located. The FDA public inspection database is non-comprehensive and excludes pre-approval inspections, so obtain firm and FDA records rather than treating this as “no inspections.”^[4] |

| FDA warning letter/import alert | No exact-name HLB PEP/AnyGen warning letter or named import-alert entry was located in the searches performed. This is a desk-search result, not regulator certification. |

| EMA/EudraGMDP non-compliance | No qualifying public record located. |

| EDQM CEP/suspension | No qualifying CEP or adverse record located. |

| PMDA/MHLW adverse action | No qualifying public adverse record located; foreign-manufacturer recognition is reported by the issuer.^[1] |

| NMPA, TGA, Health Canada, WHO | No qualifying public manufacturing approval or adverse record located. |

| Recalls | No regulator recall tied to HLB PEP/AnyGen was located. |

10. Supply-chain and operational dependencies

The audited 2023 filing identifies purchased input classes as ****amino acids, resin, synthesis solvents and purification solvents****. Purchases were booked as domestic, with named Korean vendors including Daejung Chemicals & Metals, LeeHyo Bioscience, Gwangju Science, Duksan Pure Chemicals, BeadTech and Interpharm.^[12] “Domestic purchase” does not establish domestic origin: protected amino acids, SPPS resins, HPLC media and specialty reagents may be imported upstream. Country of origin, manufacturer, broker chain, dual sourcing, change notification, GMP/compendial grade and audit status are not disclosed.

| Dependency | Desk assessment |

| --- | --- |

| Protected amino acids / specialty building blocks | Critical and origin undisclosed. Request actual manufacturer, synthesis route, chiral/impurity controls, TSE/BSE status, nitrosamine risk and dual-source position. |

| SPPS resin | Critical; resin purchases disclosed but grade, manufacturer, linker chemistry, residuals/leachables and dual sourcing are unknown.^[12] |

| Synthesis and purification solvents | Material-volume dependency; domestic distributors are named, but upstream origin, recovery/reuse and shortage plans are unknown.^[12] |

| Preparative HPLC media | Website identifies ODS C18 10 µm; vendor, lifecycle, sanitisation, column packing/repacking qualification and supply resilience are not public.^[6] |

| Single-site/product dependency | Two GMP sites reduce entity-wide concentration, but each proposed API may still be single-site. DMF-to-site mapping and validated alternate-site status are unknown. Jangseong sanction demonstrates product/site interruption risk. |

| Lyophilisation | API isolation appears dependent on classified tray lyophilisation; equipment redundancy, condenser/ice-capacity bottlenecks and backup power are unknown.^[6] |

| Cold chain | Peptide API storage/shipping temperatures, excursion limits, lane qualification, data loggers and stability support are not public. Do not assume ambient shipment. |

| Sterile fill and finished dose | Not evidenced internally; any South African sterile dosage form will depend on a separately qualified fill-finish partner. Identity, quality agreement and release chain are unknown. |

| Device / primary container / packaging | No capability evidenced; separate syringe/vial/stopper/device and packaging suppliers would be dependencies. |

| Subcontracting | No complete subcontractor register or outsourced-test list is public. Release testing, microbiology, stability storage, calibration, logistics and waste/solvent handling must be mapped. |

| Research/GMP segregation | Gwangju has disclosed non-GMP production while the entity also sells catalogue/custom and cosmetic peptides. Personnel, material, equipment, data and document segregation require audit.^{[7][12]} |

| Financial/continuity | 2025 sales were KRW 4.96 billion with an operating loss of KRW 7.32 billion;

HLB-group financing and 2026 expansion may help, but supplier should provide going-concern, capacity-funding and business-continuity evidence.[^13] |
| Geopolitical/logistics | South Korea is generally accessible, but imported upstream reagents, maritime/airfreight routes, export controls, FX exposure and regional security remain unquantified. A South Africa-specific lane and customs plan is needed. |

11. South African therapeutic-supply fit

HLB PEP may be considered for ****non-sterile peptide API/drug substance**** supplied to a separately qualified finished-dose manufacturer. Its active FDA DMFs and Korean regulatory history are useful dossier assets, but none proves acceptability to SAHPRA. There is no public evidence of a South African registration, SAHPRA inspection, local authorised representative, local product label, or established South African supply lane. Accordingly, the manufacturer must not be described as SAHPRA-approved, fully compliant, or safe.

Before any inclusion in a South African dossier or supply agreement, obtain:

- the current MFDS manufacturing licence and GMP certificates with English translations, site addresses, dosage-form/API scope and expiry;
- product authorisation/DMF documents and letters of access, with exact legal-name and site continuity from AnyGen to HLB PEP;
- a South Africa-ready API dossier (CTD Module 3/ASMF or equivalent), GMP evidence acceptable to SAHPRA, and commitments for change notification and regulator access;
- three recent commercial-scale batch records and CoAs, process/cleaning/analytical validation summaries, stability and retest

support, impurity qualification, elemental-impurity and nitrosamine risk assessments;

- 2025 sanction CAPA closure and all inspection history;
- quality/technical agreement defining release, deviations/OOS, complaints/recalls, pharmacopoeial changes, change control, transport, subcontractors and right to audit;
- GDP-qualified storage/shipping validation to South Africa, with excursion handling and import/customs responsibility;
- separate qualification of sterile fill-finish, container closure, device and finished packaging where applicable.

12. Tier rationale

****Preliminary Tier B — credible but material gaps.**** Positive signals are two currently reported MFDS-GMP plants, multiple Korean peptide API authorisations, Japanese foreign-manufacturer recognition, Indian product linkage, and three active FDA Type II DMFs. Constraints preventing Tier A are the confirmed 2025 APQR-related manufacturing suspension; absence of public CAPA closure; no public FDA inspection classification/EIR; no located EMA/EudraGMDP, EDQM CEP, TGA, Health Canada, WHO or South African evidence; incomplete product-to-site mapping; marketing rather than independent substantiation for multi-kilogram scale and validation; and major unknowns in data integrity, OOS/OOT/CAPA, stability, impurity control, subcontracting and upstream origin.

Tier B means ****suitable to receive an RFI and potentially progress to audit****. It is not a purchase approval.

13. Source assessment and caveats

Primary weight was given to FDA, MFDS and Korean exchange filings. Company webpages were used for technical detail but are labelled as claims where no regulator or audit document corroborates them. The FDA DMF spreadsheet was downloaded and searched under both “ANYGEN” and “HLB PEP.” Negative regulator-search findings are bounded by database coverage, name changes, non-public records and dynamic systems. The dossier does not infer unnamed

customers or convert applications, mock inspections, DMFs, foreign-manufacturer recognition, ISO certificates or GMP claims into product approvals.

- [^1]: Korea Exchange (KIND), HLB PEP Q1 2026 report, filed 15 May 2026, legal identity, ownership, history, MFDS GMP and product milestones: <https://kind.krx.co.kr/external/2026/05/15/002059/20260515004562/11013.htm>
- [^2]: US FDA, ****List of Drug Master Files****, Q2 2026 list current through 30 June 2026, page and official XLS download: <https://www.fda.gov/drugs/drug-master-files-dmfs/list-drug-master-files-dmfs> and <https://www.fda.gov/media/192069/download?attachment>
- [^3]: HLB PEP, notice concerning Gwangju Regional MFDS administrative disposition, 15 September 2025: <https://hlbpep.co.kr/kor/customer/notice.php?m=v&idx=1606>
- [^4]: US FDA, ****Inspection Classification Database****, limitations and dashboard coverage: <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/inspection-classification-database>
- [^5]: HLB PEP, company-carried report on private Interchem mock inspection at Osong, 24 February 2025: <https://hlbpep.co.kr/kor/customer/notice.php?m=v&idx=1548&pNo=1&code=notice>
- [^6]: HLB PEP, ****Manufacture of peptide APIs according to GMP regulations****, synthesis, purification, isolation and testing claims: <https://hlbpep.co.kr/eng/peptides/gmp.php>
- [^7]: HLB PEP homepage, legal footer and explicit research-use catalogue/service categories: <http://hlbpep.co.kr/>
- [^8]: Korea Exchange (KIND), HLB Genex beneficial-ownership report for HLB PEP, 10 July 2026: <https://kind.krx.co.kr/external/2026/07/10/000747/20260710001647/00636.htm>
- [^9]: US FDA, SRO issuer list detail for HLB Pep Co., Ltd., updated 7 August 2026: <https://www.accessdata.fda.gov/scripts/sda/sdDetailNavigation.cfm?sd=srolist&id=587C873A8BC53E9EE06346F0E10A144F&rownum=1160>
- [^10]: HLB PEP, official locations page: <https://hlbpep.co.kr/kor/company/location.php>
- [^11]: HLB PEP, company-carried profile of Jangseong and Osong operations, 15 June 2023: <https://hlbpep.co.kr/kor/customer/notice.php?m=v&idx=1398&pNo=2&code=notice>
- [^12]: Korea Exchange (KIND), AnyGen 2023 annual report, filed 20 March 2024, products, capacity, raw materials, suppliers and services: <https://kind.krx.co.kr/external/2024/03/20/000736/20240320003693/11011.htm>
- [^13]: Korea Exchange (KIND), HLB PEP 2026 shareholder-meeting notice, business, manufacturing strategy and financial statements: <https://kind.krx.co.kr/external/2026/03/12/001283/20260312003157/00591.htm>
- [^14]: HLB PEP, generic peptide portfolio and company-reported pharmacopeial/DMF descriptors: <https://hlbpep.co.kr/eng/peptides/generic.php>
- [^15]: HLB PEP, official corporate history and regulatory milestones: <http://hlbpep.co.kr/kor/company/history.php>
- [^16]: Korean Ministry of Food and Drug Safety, AnyGen product/export-country page, registered 20 May 2016: https://www.mfds.go.kr/eng/brd/m_19/view.do?seq=70331&srchFr=&srchTo=&srchWord=&srchTp=&itm_seq_1=0&itm_seq_2=0&multi_itm_seq=0&company_cd=&company_nm=&page=4
- [^17]: Korea Exchange (KIND), HLB Genex ownership report documenting largest-shareholder status in April 2025: <https://kind.krx.co.kr/external/2025/04/14/000262/20250414000741/00636.htm>

10-M10.md

M10 — ScinoPharm Taiwan, Ltd. (Taiwan)

****Decision date:** 20 August 2026**

****Purpose:** Preliminary desk prequalification for regulated therapeutic-peptide supply to a South African business**

****Preliminary desk tier:**** ****A** — strong public regulated-market evidence; suitable to advance to a formal RFI and risk-based audit, not approved for supply******

1. Executive decision

ScinoPharm Taiwan, Ltd. is a Taiwanese listed pharmaceutical manufacturer with unusually strong public evidence for a peptide supplier: an active Taiwanese API/GDP listing that names multiple peptide or peptide-like drug substances, a Taiwanese sterile-dosage-form GMP listing, an FDA-recognised U.S. finished-product label for glatiramer acetate injection under ANDA 214741, a long FDA inspection history, and audited financial statements identifying the group, its ultimate parent and its China subsidiary.[1][2][3][4][5] The evidence supports advancement to an RFI for **peptide API/drug substance and, subject to product-specific confirmation, sterile injectable fill-finish**.

The decision is not unconditional. Public evidence does not expose the current FDA final classification/EIR for the 2022 or March 2024 sterile-site inspections; company statements of “zero deficiencies” are not substitutes for FDA classification letters. A third-party regulatory record reports a **2012 FDA VAI** inspection, and FDA’s own FOIA log indexes a 2015 Form 483 and a 2015 EIR for the Taiwan site; the underlying records were not available in this review.[6][7][8] Public sources also do not adequately disclose peptide-specific OOS/OOT/CAPA performance, data-integrity governance, protected-amino-acid/resin/solvent sources, supplier qualification, impurity purge strategies, subcontractors, or business-continuity stocks. These are mandatory RFI and audit topics.

No public evidence located by the decision date demonstrated a SAHPRA registration, South African product label, or South African GMP reliance decision naming ScinoPharm. Therefore, this dossier **does not state or imply SAHPRA approval, full compliance, product safety, or readiness to release product into South Africa**.

2. Legal entity, ownership and group perimeter

[illegible]

ScinoPharm is listed on the Taiwan Stock Exchange under code ****1789****. Uni-President Enterprises Corp. held 37.94% at 31 March 2026, while the National Development Fund held 13.85%.[11] More importantly, the audited 2025 consolidated statements expressly identify ****Uni-President Enterprises Corp. as the ultimate parent and ultimate controlling party****. [5] Thus, ScinoPharm is a separately listed legal entity within the Uni-President-controlled group, not a wholly owned Uni-President subsidiary.

The audited filing identifies ****SciAnda (Changshu) Pharmaceuticals, Ltd.**** as a 100%-owned indirect Mainland China subsidiary engaged in API/new-drug R&D and manufacture. It also identifies ****SciAnda Shanghai Biochemical Technology, Ltd.**** as a 100%-owned import/export and

API/intermediate sales entity, and non-manufacturing investment entities SPT International, Ltd. and ScinoPharm Singapore Pte Ltd.[5] The company's official global-presence page gives the Changshu manufacturing address and distinguishes the Shanghai entity and Japan office.[12]

3. Sites and roles

Site / entity	Publicly evidenced role	Therapeutic-supply relevance and boundary

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| ****ScinoPharm Taiwan, Ltd., No. 1 Nan-Ke 8th Road, Shan-Hua, Tainan 741014, Taiwan**** | Registered operating pharmaceutical factory; headquarters; API R&D and manufacturing; peptide API; sterile injectable formulation/fill-finish; testing; warehousing/GDP, including cold-chain medicines.[1][2][9][10][13] | ****Primary qualifying site.**** Product- and building-specific scope must be fixed in the quality agreement. Public records place API and finished dose at the same address but do not prove every peptide is made and filled in the same building or campaign. |

| ****SciAnda (Changshu) Pharmaceuticals, Ltd., No. 16 Dong Zhou Road, Economic Development Zone, Changshu, Jiangsu 215513, China**** | 100%-owned indirect subsidiary; API/new-drug R&D and manufacture. Company lists semaglutide full side chain as a China-produced peptide-related API/intermediate and states the Changshu API site shares its quality system with Taiwan.[5][12][14][15] | ****Relevant only where the proposed route/product uses Changshu.**** It is not evidence that all peptide APIs are dual-sited. Its inclusion creates China regulatory, trade, logistics and cross-strait dependencies. |

| ****ScinoPharm R&D laboratory, Nangang National Biotechnology Research Park, Taipei**** | Company-described R&D laboratory.[15] | ****Development/research only on current public evidence.**** No therapeutic GMP production qualification should be attributed without an explicit licence and project scope. |

| ****SciAnda Shanghai Biochemical Technology, Ltd., Shanghai**** | Import/export and sale of APIs/intermediates; official address in Shanghai.[5][12] | ****Commercial/logistics entity, not evidenced as a therapeutic manufacturing site.**** |

| ****Japan office, Tokyo**** | Commercial office.[12] | ****No manufacturing role evidenced.**** |

4. Capability boundary

4.1 Peptide API / drug substance

The company markets SPPS, LPPS, hybrid synthesis, PEG platforms, segment condensation and copolymer approaches, with tangential-flow filtration and DAC/MCSPG column chromatography. It claims peptide capacity up to ****200 kg/year****, while its broader API estate has 18 lines and 20–10,000 L reactors.[15] These are ****manufacturer claims****, not regulator-certified nameplate capacity. A buyer should request actual peptide train sizes, maximum demonstrated batch size and annual throughput by molecule class, reactor/synthesiser occupancy, downstream bottlenecks, yield history and campaign-changeover data.

The official commercial list identifies chemically synthesised peptide products and production sites: ****bivalirudin, glatiramer acetate, liraglutide, octreotide acetate and teriparatide in Taiwan; semaglutide full side chain in China****. It states U.S. regulatory submissions for each.[14] TFDA's current domestic API list independently names, at the Tainan address, ****desmopressin acetate, glatiramer acetate, leuprolide mesylate, liraglutide and teriparatide****, among other APIs.[1] The glatiramer evidence is especially material: DailyMed records an FDA-regulated finished product under ANDA 214741, and ScinoPharm states it developed the process and secured FDA approval.[3][16]

4.2 Sterile finished dose and fill-finish

TFDA lists the Tainan site for aseptically prepared small-volume liquid injections and, for cytotoxic/high-toxicity products, aseptically prepared lyophilised injections and terminally sterilised

small-volume liquid injections. TFDA also lists storage, supply and export, including cold-chain medicines.[2] This is regulator evidence of sterile finished-dose scope, not merely marketing.

The company describes two independent lines: one for prefilled syringes and cartridges, up to 7,800 one-mL units/hour, and a vial line with a 10 m² lyophiliser and an approximate maximum batch of 19,000 10R vials. It describes closed isolators, automated filling, and single-use sterile filtration, transfer tubing and hold bags.[15] The CDMO page adds sterile liquids, lyophilised injectables, syringes

and cartridges.[13] Capacity, intervention design, media-fill history, container-closure integrity and peptide-specific line experience remain to be confirmed in an RFI/audit.

4.3 Device, assembly and packaging

ScinoPharm markets injection-pen assembly, secondary packaging, serialisation and automatic assembly/packaging; it also states that it works with pen-injector and autoinjector supply networks.[13][15] These statements support a potential ****combination-product assembly and packaging service****, but do not establish that ScinoPharm manufactures device components. The U.S. glatiramer label is for glass prefilled syringes and names ****Upsher-Smith Laboratories, LLC as packager/distributor****; it does not publicly identify every component manufacturer or attribute autoinjector manufacture to ScinoPharm.[3] Device-component, needle, syringe-barrel, stopper/plunger, safety-device and external packager responsibilities must therefore be mapped contractually.

4.4 Research-only services

ScinoPharm expressly offers milligram-to-gram laboratory-use peptide synthesis, alongside preclinical/toxicology batches, clinical GMP API and commercial GMP API.[17] The mg-g laboratory service is ****research-only unless a batch is separately manufactured, tested, released and documented under an applicable therapeutic GMP scope****. Research-use materials and grey-market peptide listings must be excluded from any therapeutic supply decision.

5. Regulated-market and product evidence

| Market / authority | Evidence | Interpretation |

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| ****United States / FDA**** | DailyMed lists glatiramer acetate injection, 20 and 40 mg/mL prefilled syringes, ANDA 214741, marketing start 23 January 2026, packaged/distributed by Upsher-Smith.[3] Company also reports U.S. ANDA approval for internally developed/manufactured bortezomib injection in September 2023.[18] | Strong finished-product evidence in the United States. The glatiramer label evidences a regulated product and dosage form, but public label text does not expose the complete manufacturing chain. |

| ****Taiwan / TFDA**** | Current API list names peptide substances at Tainan; current sterile-facility listing identifies aseptic/terminal sterilisation and cytotoxic lyophilised scope; company reports TFDA marketing approval for bortezomib injection in January 2026.[1][2][19] | Strong domestic regulator evidence for site scope. Bortezomib is not a peptide, but validates sterile finished-dose execution. |

| ****European Union / EDQM/EudraGMDP**** | EDQM CEP R1-CEP 2016-002 Rev 00, renewed 2022, names ScinoPharm as holder for ****exemestane****, a non-peptide steroid.[20] An archived Hungarian GMP certificate records a 2008 Tainan API inspection for gemcitabine/irinotecan and warns not to rely on it after three years.[21] Company claims a 2013 EMA and 2016 EDQM site inspection and commercial peptide support to Europe.[22][17] | Evidence of European API regulatory history, but the available CEP and old certificate are ****not peptide-specific**** and the 2008 certificate is stale. No current peptide-specific EudraGMDP certificate was retrieved. |

| ****Japan / PMDA-MHLW**** | Company history reports Taiwan PMDA inspections in 2008 and 2017 and Changshu inspection in 2018. A PMDA foreign-manufacturer public workbook was located with an indexed ScinoPharm Tainan entry, but its structured fields were not reliably extractable during

this review.[22][23] | Credible presence, but current accreditation category/expiry and product linkage require documentary confirmation. |

| **Brazil / ANVISA** | Company reports an API-site inspection on 11–15 November 2024 with zero defects and says ten APIs were registered through customers.[24] | Useful emerging-market signal; regulator-hosted certificate/inspection report was not retrieved, and the statement is not peptide-specific. |

| **China / NMPA** | Company reports a 2020 NMPA registration/GMP inspection at Changshu.[22] | Company claim only in this review; produc

t-specific China authorisations should be obtained. |

| **Australia / TGA; Canada; Korea / MFDS; WHO** | Company reports a 2007 TGA and 2008 Korean inspection and lists non-peptide submissions in Australia/Canada; no current primary regulator record linking therapeutic peptide manufacture was retrieved.[14][22] | **Peptide market service not publicly established on the reviewed record.** Do not infer approval or current supply. |

| **South Africa / SAHPRA** | No ScinoPharm-named public registration, label or GMP reliance record was located in the searches performed to the decision date. | **Not publicly disclosed / not evidenced.** South African dossier, reliance, import, licensing and pharmacopoeial requirements remain sponsor responsibilities. |

The audited filing corroborates actual group sales in **USA, Europe, India, Taiwan and other Asian markets** in 2025, but does not allocate those revenues to peptides or disclose customers. It reports 2025 revenue of NTD 855.948 million in the USA, 479.487 million in Europe, 438.401 million in India, 400.925 million in Taiwan and 950.463 million in Asia (as presented in the filing).[5] These figures evidence geographic service, not customer identity or peptide approval. **No customer has been inferred.**

6. Regulatory inspection and adverse-history review

ScinoPharm's history lists Taiwan-site FDA inspections beginning in 2001 and continuing through 2005, 2008, 2012, 2015, 2017, 2019, twice in 2022, and March 2024. It describes the 2022 and 2024 sterile-site inspections as zero-observation/zero-deficiency outcomes.[8][18][22] These later outcomes are company announcements; a current FDA final classification letter or EIR should be requested.

FDA's public database explains that NAI, VAI and OAI are **final project-area classifications**, that its dashboard is not a comprehensive list, and that pre-approval inspections are excluded.[25] That limitation is important because ScinoPharm's sterile inspections were described as pre-approval inspections.

Adverse or cautionary findings located were:

1. A third-party mirror of FDA inspection data identifies **inspection 778186, 28 March 2012, drugs/CDER, as VAI** in its search result. The fetched detail confirms the inspection identifier/date/product area but did not expose the classification field; obtain the FDA classification letter/EIR before relying on the classification.[6]
2. FDA's FY2015 FOIA log indexes a request for a **ScinoPharm Taiwan Form 483**, while its FY2016 log indexes a ScinoPharm EIR for the March–April 2015 period. A commercial document index separately identifies an EIR dated 27 March 2015.[7] The observations, response, CAPA and final classification were not reviewed.
3. A third-party mirror reports **2019 inspection 1094253 as NAI**; the fetched detail confirms the May 2019 drugs/CDER inspection but not the classification field.[26]
4. Searches of FDA warning-letter, import-alert and recall sources did not locate a ScinoPharm-named warning letter, import alert or recall by the decision date. This is **not proof of absence**, particularly because public databases are incomplete and product recalls can be filed under an application holder/distributor.
5. No EudraGMDP non-compliance statement, WHO notice, TGA action, Health Canada action, MFDS action, NMPA action or TFDA serious GMP action naming ScinoPharm was located in this

review. Again, this is a search result, not an assurance.

The company's use of "zero Form 483" is more precise than "passed FDA inspection," but neither phrase establishes a final NAI classification. FDA itself explains that classification follows agency review and can differ from the investigator's close-out.[25]

7. Analytical, impurity and quality-system evidence

Verified or regulator-corroborated

TFDA's API and sterile-site listings demonstrate licenced scope at the Tainan address, including named peptide substances, sterile products and cold-chain GDP operations.[1][2] FDA's

DailyMed label verifies the approved glatiramer strengths, prefilled-syringe presentation and the product's U.S. application number.[3] The 2025 consolidated statements received an unmodified audit opinion; this supports financial-reporting reliability and corporate continuity, but **is not a GMP audit**.[5]

Manufacturer claims requiring qualification

ScinoPharm states that peptide development includes process and analytical-method development/transfer, CPP/NOR/POR characterisation and analytical validation.[17] It claims in-house peptide drug-product testing, peptide-stability experience, physicochemical characterisation, formulation optimisation, freeze-thaw and thermal studies, lyophilisation cycle development/transfer/scale-up, and regulatory support.[15] For glatiramer, it states that more than 40 analytical techniques, statistical modelling and biological-activity assays were used to establish equivalence.[16]

The company's contract-testing list includes nitrosamines by high-resolution mass spectrometry, elemental impurities, XRPD, NMR, SEM, DSC, EDS and DVS. Extractables/leachables were still marked **"under preparation"**, so this should not be treated as an established in-house capability.[27] Its public quality page describes QRM/QbD, an electronic QMS with traceability, data analytics for deviations/process monitoring, training and quality metrics.[28]

Material evidence gaps

Public sources do not provide peptide-specific specifications or validated release/stability method lists. The RFI must obtain, for the proposed API and drug product: identity and assay methods; related-substance and deletion/insertion/truncation sequence controls; stereoisomer and racemisation control; residual solvents; water/counter-ion; residual reagents and scavengers; resin/linker-derived impurities; aggregation/oligomerisation; elemental impurities; bioburden/endotoxin where applicable; particle/subvisible particle methods; container-closure integrity; microbial/sterility methods; reference-standard qualification; method-validation summaries; forced degradation; stability protocols and actual trends.

No public, product-specific metrics were located for **data integrity, audit-trail review, OOS, OOT, deviation ageing, CAPA effectiveness, invalidated assays, laboratory incidents, repeat testing or annual product review**.

The eQMS claim is not enough. Request three years of metrics and representative, redacted investigations, including the 2012 VAI and 2015 Form 483 remediation history.

8. Supply-chain and dependency map

| Dependency | Evidence and risk | Required mitigation |

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| **Protected amino acids, resins/linkers, coupling/deprotection reagents, solvents and chromatography media** | These inputs are inherent to the claimed SPPS/LPPS and

chromatography processes, but ScinoPharm does not publicly identify qualified sources, countries of origin, dual sourcing, incoming tests or safety stocks.[15] | Product-specific bill of materials; manufacturer and country for critical inputs; animal/TSE/BSE and allergen statements; change-notification rights; dual-source status; incoming-ID strategy; nitrosamine/genotoxic risk assessment; solvent recovery controls. |

| ****Taiwan single-site concentration**** | Most listed peptide APIs and all evidenced sterile fill-finish are assigned to Tainan.[1][2][14] Public evidence does not show a qualified backup site for liraglutide, glatiramer, octreotide, bivalirudin or teriparatide. | Business-continuity plan; utilities and disaster recovery; inventory coverage; alternate analytical laboratory; technology-transfer package; regulatory strategy for a second source/site. |

| ****China / Changshu**** | Semaglutide full side chain is listed at the China site; Changshu is a 100%-owned API manufacturer and conducted substantial intercompany supply to Taiwan in 2025.[5][14] | Route-specific genealogy; confirm whether any South Africa product uses Chinese intermediates; sanctions/expor

t-control and customs review; cross-strait contingency; separate audit and quality agreement for Changshu. |

| ****Sterile-fill and lyophilisation bottleneck**** | One marketed PFS/cartridge line and one vial/lyophiliser line are described. Public data do not establish redundant fill lines or lyophilisers.[15] | Line allocation, surge capacity, maintenance/spares, media-fill and sterility-assurance history, alternate fill site, maximum hold times and bulk-shipping option. |

| ****Device/components**** | PFS, cartridge and pen assembly are offered, but component manufacturers are undisclosed; the U.S. label names an external packager/distributor.[3][13] | Full component/subcontractor map, supplier quality agreements, ISO 11608/ISO 11040 applicability, human-factors responsibilities, E&L/leachables plan, compatibility and functionality trends. |

| ****Cold chain and logistics**** | TFDA authorises storage/supply/export including cold-chain medicines; glatiramer label requires refrigerated storage with limited room-temperature exposure.[2][3] | Lane qualification to South Africa, validated shipper, excursion decision tree, data loggers, customs-delay study, local qualified importer/warehouse and returns/destruction process. |

| ****Subcontracting**** | Public sources acknowledge external device supply networks and a U.S. packager/distributor; other testing, sterilisation, microbiology, transport and waste subcontractors are not disclosed.[3][15] | Approved-subcontractor list, operations and countries, audit status, data ownership, onward-subcontracting prohibition/consent, change control and right to audit. |

| ****Inventory and long lead times**** | Auditors identified complex manufacturing, long material-preparation lead times and uncertain registration timing as drivers of material inventory-valuation risk.[5] | Forecast model, frozen horizon, safety-stock agreement, shelf-life at dispatch, capacity reservation, obsolete-stock allocation and launch-delay terms. |

| ****Geopolitical / natural-hazard exposure**** | Taiwan and Jiangsu concentration creates cross-strait, shipping-lane and regional disruption exposure. This is a risk inference, not evidence of a present disruption. | Scenario-tested continuity plan, dual ports/forwarders, cyber and power recovery, earthquake/typhoon response, political-risk and cargo insurance. |

9. South Africa fit and prequalification conditions

ScinoPharm is a ****credible RFI candidate**** for South African regulated therapeutic supply because the core Tainan legal entity and factory are readily resolved, TFDA provides current API and sterile-scope evidence, the U.S. has an FDA-regulated complex polypeptide finished product under an ANDA, and audited filings show a functioning API/injection business with sales across regulated markets.[1][2][3][5]

Before any technical or commercial nomination, require:

1. Current Tainan and, if relevant, Changshu manufacturing authorisations, TFDA PIC/S GMP evidence and inspection scope; FDA FEI numbers and final classification letters/EIRs for 2019, 2022 and 2024; the 2012 and 2015 Form 483/EIR responses and CAPA closure evidence; current EU/EDQM, PMDA/MHLW, NMPA and ANVISA documents relevant to the exact product.
2. Product-specific regulatory history: DMF/ASMF numbers and status, letters of access, countries

3. A complete site/process map from starting materials through peptide API, sterile bulk, fill, inspection, device assembly, labelling, serialisation, release, storage and shipment, with every subcontractor disclosed.

5. Quality-system evidence: data-integrity assessment, audit-trail governance,

6. Sterility-assurance package where fill-finish is proposed: contamination-control strategy, isolator qualification, APS/media fills, environmental and personnel monitoring trends, sterilisation/depyrogenation, filter validation, hold times, CCIT, visual inspection, particulate and E&L data.

10. Preliminary tier rationale

References

[7]: U.S. FDA, FY2015 FOIA Log, <https://www.fda.gov/media/184915/download> ; U.S. FDA,

FY2016 FOIA Log, <https://www.fda.gov/media/184916/download>

[8]: ScinoPharm Taiwan, Ltd., "Both Injectable Plant and China Subsidiary Pass US FDA Inspection with Zero Deficiencies," 6 May 2024, <https://www.scinopharm.com/news-detail/115/>

[9]: Taiwan Ministry of Economic Affairs, Business Registration, unified business number 16130161, <https://findbiz.nat.gov.tw/fts/company/16130161>

[10]: Taiwan Ministry of Economic Affairs, Factory Registration 94A00015, <https://findbiz.nat.gov.tw/fts/factory/08/94A00015>

[11]: ScinoPharm Taiwan, Ltd., "Major Shareholders," as of 31 March 2026, <https://www.scinopharm.com/investor-detail/major-shareholders/>

[12]: ScinoPharm Taiwan, Ltd., "Global Presence," <https://www.scinopharm.com/brand/20/>

[13]: ScinoPharm Taiwan, Ltd., "Sterile Injectables Contract Manufacturing," <https://www.scinopharm.com/products-detail/56/>

[14]: ScinoPharm Taiwan, Ltd., "Commercial Products," accessed for decision date 20 August 2026, <https://www.scinopharm.com/products-detail/commercialAPI/>

[15]: ScinoPharm Taiwan, Ltd., "CMC Capabilities," <https://www.scinopharm.com/solutions/SPTCMC/>

[16]: ScinoPharm Taiwan, Ltd., "ScinoPharm Secures U.S. FDA Approval for Glatiramer Acetate Injection," 5 January 2026, <https://www.scinopharm.com/news-detail/137/>

[17]: ScinoPharm Taiwan, Ltd., "Peptide CDMO Service," <https://www.scinopharm.com/products-detail/54/>

[18]: ScinoPharm Taiwan, Ltd., "ScinoPharm Taiwan obtains Bortezomib for Injection ANDA approval," 2 October 2023, <https://www.scinopharm.com/news-detail/117/>

[19]: ScinoPharm Taiwan, Ltd., "Approval of Bortezomib Injection in Taiwan," 21 January 2026, <https://www.scinopharm.com/news-detail/139/>

[20]: European Directorate for the Quality of Medicines & HealthCare, CEP R1-CEP 2016-002 Rev 00, Exemestane, renewed 6 October 2022, copy hosted by certificate holder, <https://www.scinopharm.com/download/57/>

[21]: Hungarian National Institute of Pharmacy, GMP Certificate HUN-37451-68-2007, issued 2 June 2008, archived copy hosted by ScinoPharm, <http://jp.scinopharm.com/upload/download/files/220170913154210.PDF>

[22]: ScinoPharm Taiwan, Ltd., "Company / Milestones," <https://www.scinopharm.com/about/>

[23]: Pharmaceuticals and Medical Devices Agency, Japan, public foreign-manufacturer listing workbook, <https://www.pmda.go.jp/files/000240884.xlsx>

[24]: ScinoPharm Taiwan, Ltd., "Zero-Deficiency Record in ANVISA Inspection," 20 November 2024, <https://www.scinopharm.com/news-detail/113/>

[25]: U.S. FDA, "Inspection Classification Database," content current 13 September 2024, <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/inspection-classification-database>

[26]: KeyPedia/Global Key Solutions, "FDA Inspection 1094253 — Scinopharm Taiwan Ltd — May 17, 2019" (secondary mirror), https://www.globalkeysolutions.net/records/fda_inspections/scinopharm-taiwan-ltd/7044bf7c-e911-4e7c-84ce-056f5f8b6169

[27]: ScinoPharm Taiwan, Ltd., "Contract Analytical Testing Services," <https://www.scinopharm.com/products-detail/60/>

[28]: ScinoPharm Taiwan, Ltd., "Quality Standards and Management," <https://www.scinopharm.com/solutions/QC/>

[29]: ScinoPharm Taiwan, Ltd., "Summary of ScinoPharm Taiwan, Ltd. Submissions," updated 1 June 2026, <https://www.scinopharm.com/download/80/>

****Desk-review limitation:**** This dossier uses public records available or located as of the decision date. Non-public DMFs/ASMFs, EIRs, inspection classification letters, Establishment Inspection Reports, quality agreements, audit reports, batch data and customer dossiers were not reviewed. Database non-hits are not proof that an event or record does not exist.